

Integral cryptanalysis in characteristic p

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Abstract. Integral and ultrametric integral cryptanalysis are generalized to finite rings of prime characteristic p that are isomorphic to a product of fields. This extends, for instance, the complete state of the art in integral cryptanalysis from $\mathbf{F}_2^n$ to $\mathbf{F}_q^n$, for all prime powers q . A compact representation of transition matrices, based on convex polyhedra, is introduced to ensure that the proposed methods are computationally efficient even for large p . Automated tools are developed and applied to a few generic and several concrete primitives. The analysis shows that previous degree estimates for Feistel-GMiMC, HadesMiMC, AES-prime, small-pSquare and mid-pSquare are overly optimistic. Furthermore, except for AES-prime, these primitives do not meet their design criteria unless their number of rounds is increased.

Keywords: Geometric approach · Integral cryptanalysis · Ultrametric integral cryptanalysis · Arithmetization-oriented primitives

1 Introduction

Symmetric-key primitives are increasingly designed for nontraditional applications such as multiparty computation, zero-knowledge proofs, and homomorphic encryption. Primitives optimized for these use-cases are defined in terms of arithmetic operations in diverse algebraic structures, including large extension fields (MiMC, GMiMC, ... [1, 2]), small prime-order fields (small-pSquare, AES-prime, ... [23, 33]), and large prime-order fields (Poseidon, Rescue, ... [3, 21]).

The security analysis of these *arithmetization-oriented primitives* is not yet mature. Traditional methods such as linear and differential cryptanalysis are curbed by the excellent properties of low-degree nonlinear functions over finite fields. Especially for large fields, algebraic attacks seem to be the only game in town. To develop security arguments against algebraic methods such as Gröbner bases, interpolation and zero-sums, a good understanding of the polynomial representation of the primitive is necessary.

For primitives defined over $\mathbf{F}_2^n$, the introduction of the division property [24, 37, 38], parity sets [16], monomial prediction [25] and algebraic transition matrices [14] have turned integral cryptanalysis into a well-developed theory that can be used to determine degrees, zero-sum properties, etc. For all other algebraic structures — except to some extent for $\mathbf{F}_{2^n}$ [18] — integral cryptanalysis

is surprisingly underdeveloped, and designers consequently rely on ad-hoc methods to determine the algebraic properties of their primitives. In contrast, linear and differential cryptanalysis are well-understood for arbitrary finite Abelian groups [5, 7, 9].

Previous work. Attempts to generalize integral cryptanalysis have mainly focused on extension fields of $\mathbf{F}_2$, starting from the observation that the (univariate) degree of a primitive on $\mathbf{F}_{2^n}$ is related to its (multivariate, or *algebraic*) degree as a function on $\mathbf{F}_2^n$. For example, at Asiacrypt 2020, Eichlseder et al. [20] used this to show that the algebraic degree of MiMC grows linearly with the number of rounds. A more detailed analysis of MiMC is given by Bouvier, Canteaut and Perrin [17]. Yet another specialized approach to bounding algebraic degrees is the coefficient grouping method of Liu et al. [31, 32]. Finally, at Asiacrypt 2022, Cui, Hu, Wang and Wei [18] generalized monomial prediction to $\mathbf{F}_{2^n}^m$.

For finite fields other than $\mathbf{F}_{2^n}$, relatively little is known. At Crypto 2020 [12], higher-order differential properties were obtained for GMiMC and HadesMiMC over $\mathbf{F}_q^n$, with q either a large prime or a high power of two. However, the degree bounds used to obtain these properties are naive. A more interesting observation from [12] is that multiplicative subgroups of $\mathbf{F}_q^\times$ can be exploited to reduce the data-complexity.

At Asiacrypt 2024 [15], a generalization of integral cryptanalysis called *ultrametric integral cryptanalysis* was introduced. It is a multiplicative analogue of linear cryptanalysis, meaning that the underlying theoretical framework is the same but the central role of addition (exclusive-or) is taken over by multiplication (bitwise-and). Ultrametric integral cryptanalysis allows for the study of divisibility properties. It is defined in characteristic zero, so that reduction modulo two yields integral cryptanalysis over $\mathbf{F}_2^n$. Hence, if a version of ultrametric integral cryptanalysis over $\mathbf{F}_q^n$ for $q \neq 2$ could be constructed, then this should also yield a theory of integral cryptanalysis over $\mathbf{F}_q^n$. From a theoretical point of view, working in characteristic zero has several advantages. For example, it allows relating the integral and linear properties of a primitive [15, §6.4]. From a practical point of view, it gives access to higher-divisibility properties.

Contribution. This paper extends ultrametric integral cryptanalysis to all finite rings of prime characteristic p , satisfying some necessary conditions. The theory has the desired ‘reduction modulo p ’, giving for instance a full generalization of integral cryptanalysis to $\mathbf{F}_q^n$ for all prime powers q . Additional techniques are introduced to turn the theoretical concepts into computationally efficient methods, even for large p . Automated tools are developed and applied to a few generic and several concrete primitives as a proof-of-concept. Our analysis shows that previous degree estimates for Feistel-GMiMC, HadesMiMC, AES-prime, small-pSquare and mid-pSquare are overly optimistic, and that — with the exception of AES-prime — their number of rounds must be increased to meet their design criteria.

The main theoretical contributions of this paper are contained in Sections 2 to 4. Theorem 3 in Section 2 classifies all finite rings that support a general-

Table 1: Number of rounds to maximize degree and to meet the design criteria.

Primitive	Maximal degree		Security argument	
	Claim	Actual	Claim	Actual
AES-prime	5	6	14	< 14
small-pSquare	7	15	36	38
mid-pSquare [†]	20	35	56	76
HadesMiMC [‡]	2	83	88	168
GMiMC [*]	42	83	84	169

[†] For tweaks of length τ , add 8τ rounds in the ‘security argument’ columns.

[‡] For 256-bit state size, 256-bit security level and 128-bit prime. Partial rounds only. Maximal degree is for the inverse function, the forward claim is 82 (actual 83).

^{*} For the two-branch Feistel instantiated with $p = 2^{64} - 2^{32} + 1$ and 128-bit security.

ization of ultrametric integral cryptanalysis that preserves the core properties of the $\mathbf{F}_2^n$ -theory: such rings must be isomorphic to a product of finite fields. The core of the theory is developed in Section 3. Unsurprisingly, there are many similarities to the $\mathbf{F}_2^n$ -theory from [15]. However, several new facets should be highlighted. Section 3.1 shows that the base field must be a cyclotomic extension of the p -adic numbers rather than the rational numbers; this leads to several other technical differences. The choice of input sets is more flexible than for $\mathbf{F}_2^n$. Theorem 4 in Section 3.4 generalizes the observation from [12] regarding multiplicative subgroups. Unlike for $\mathbf{F}_2^n$, the properties of addition are nontrivial. They are described in Theorems 8 and 9 of Section 4.

Section 5 introduces important practical methods to bound the correlations of trails, which is the main computational hurdle for applications. If p is small, then direct modelling is feasible — in fact, for $q = 2^n$ and after reducing modulo two, ultrametric integral trails coincide with the monomial trails introduced by Cui et al. [18] at Asiacrypt 2022. For large p , we propose a compact representation of the transition matrix of a function as a *Newton polyhedron*. These are convex sets in $\mathbf{R}^n \times (\mathbf{R} \cup \{\infty\})$ that generalize the parity sets of Boura and Canteaut [16]. Degree bounds, minimum data properties, ... can be computed by solving integer linear programming problems over the interior of these polyhedra. This approach is combined with an additional specific optimization for extension fields.

Sections 6 and 7 contain the applications¹. Theorem 10 in Section 6 characterizes the properties of generic degree- d functions by analyzing trails². The same method can be used to analyze isomorphisms $\mathbf{F}_q^n \rightarrow \mathbf{F}_{q^n}$ and their inverse,

¹ The code used for the analysis of these applications can be found at <https://github.com/KULeuven-COSIC/integral-cryptanalysis-characteristic-p>.

² This result could be of independent interest, as it implies the Ax-Katz theorem [27].

which generalizes the idea that the algebraic degree of a function on $\mathbf{F}_{2^n}$ is related to its univariate degree. Based on these results and the Newton polyhedra method, we analyze generic SHARK-like constructions and Feistel ciphers. For Feistel ciphers, we additionally demonstrate how the polynomial density can be estimated by counting lattice points in a polyhedron. Section 6 contains our results for concrete primitives. The block ciphers AES-prime [33] (Eurocrypt 2023), small-pSquare (Eurocrypt 2024) [23] and mid-pSquare [6] are defined over a prime-order field with $p \leq 2^{32}$ and are modelled using constraint programming. For HadesMiMC and GMiMC, we use the Newton polyhedra method. A small sample of the results is shown in Table 1.

For all of the analyzed primitives, we find that the designers overestimated the degree. Their number of rounds is determined based on security arguments against several (mostly algebraic) attacks. Our analysis shows that, for the original security arguments against interpolation attacks to hold, more rounds are necessary. For small-pSquare and mid-pSquare, our results are based on improved estimates of the polynomial density³. For HadesMiMC and GMiMC, we consider instances that target the highest security level as described in [22, §3.1]. The security arguments in [6, 22, 23, 33] are not tight, so it is unlikely that there are corresponding attacks. Since our main focus is on building the techniques needed by cryptanalysts and designers, we leave estimating the actual number of rounds that can be broken by an interpolation attack as future work. Additional results on these primitives, such as an eight-round attack on AES-prime with $\leq 2^{59}$ data and $\leq 2^{43}$ time, can be found in Sections 7.1 and 7.2.

2 Rings supporting integral cryptanalysis

This section classifies all rings that allow for the construction of a theory of (ultrametric) integral cryptanalysis with the same desirable properties as [15].

Remark 1. Technically, the construction of ultrametric integral cryptanalysis in [15] relies only on the multiplicative structure of $\mathbf{F}_2^n$ as a monoid with the bitwise-and operation. However, without additive structure, there is not much to say beyond the basic propagation rules; hence the focus on rings.

2.1 Geometric approach

Ultrametric integral cryptanalysis was developed using the geometric approach to symmetric-key cryptanalysis [9]. This framework is intended to simplify the introduction of new cryptanalytic methods, and has previously been applied to linear [7], differential [13] and integral cryptanalysis [14, 15, 26].

The geometric approach provides a dictionary between combinatorial statements and linear algebra. For example, a statement involving a finite set X becomes a statement about the free vector space $k[X]$ on X . Here, k is a field

³ All computational results presented in this work were computed in less than 10 minutes on an AMD Ryzen 7 PRO 4750G CPU with 16GB of memory.

and $k[X]$ consists of all vectors $\sum_{x \in X} u_x \delta_x$, where the coordinates u_x are elements of k and δ_x are formal basis vectors. A function $F: X \rightarrow Y$ corresponds to a linear map $T^F: k[X] \rightarrow k[Y]$ defined by $T^F(\delta_x) = \delta_{F(x)}$, the *pushforward* of F . The first two columns of Table 2 provide an overview of the correspondence obtained in this way. A formalization of this equivalence can be found in [10].

Table 2: Abridged dictionary of the geometric approach.

Classical description	Geometric approach	
	Covariant	Contravariant
Set X	Coalgebra $k[X]$	Algebra k^X
$X \times Y$	$k[X] \otimes k[Y]$	$k^X \otimes k^Y$
Function F	Pushforward T^F	Pullback $T^{F^\vee}$
$X \rightarrow Y$	$k[X] \rightarrow k[Y]$	$k^Y \rightarrow k^X$
$F_r \circ \dots \circ F_1$	$T^{F_r} \dots T^{F_1}$	$T^{F_1^\vee} \dots T^{F_r^\vee}$
$F_1 \parallel \dots \parallel F_n$	$T^{F_1} \otimes \dots \otimes T^{F_n}$	$T^{F_1^\vee} \otimes \dots \otimes T^{F_n^\vee}$

The last column of Table 2 describes another, dual, point of view. Here, a finite set X corresponds to the space k^X of k -valued functions on X . The vector space k^X is the dual space of $k[X]$. Indeed, every k -valued function on X extends to $k[X]$ by linearity. Conversely, every linear function $f: k[X] \rightarrow k$ is uniquely determined by $f(\delta_x)$ for all x in X . A function $F: X \rightarrow Y$ gives rise to a linear map $T^{F^\vee}: k^Y \rightarrow k^X$ defined by $T^{F^\vee}(f) = f \circ F$, the *pullback* of F .

The correspondence in Table 2 is well-behaved with respect to composition and parallel application of functions. In the following theorem, $F_1 \parallel \dots \parallel F_n$ is the function defined by $(x_1, \dots, x_n) \mapsto (F_1(x_1), \dots, F_n(x_n))$ and the tensor product $\otimes$ may be concretely realized as the Kronecker product of matrices.

Theorem 1. *For the pushforward of a function $F: X \rightarrow Y$,*

- (1) *If $F = F_1 \parallel \dots \parallel F_n$, then $T^F = \bigotimes_{i=1}^n T^{F_i}$.*
- (2) *If $F = F_r \circ \dots \circ F_1$, then $T^F = T^{F_r} \dots T^{F_1}$.*

Cryptanalytic properties of a function $F: X \rightarrow Y$ are pairs (u, v) with u in $k[X]$ and v in k^Y . The *evaluation* of (u, v) is the scalar $v(T^F u)$, where v is considered to be a linear function on $k[X]$ as explained above.

Example 1. For $X = \mathbf{F}_2^n$ and $k = \mathbf{F}_2$, let S be a subset of X and f a Boolean function. For $\delta_S = \sum_{x \in S} \delta_x$, the evaluation of the property (δ_S, f) is equal to $f(T^F \delta_S) = \sum_{x \in S} f(F(x))$. Properties of this type [14, Definition 5], especially with S an affine subspace and f a monomial function, are studied using integral cryptanalysis.

Most of the time, computing the exact value of $v(T^F u)$ is not feasible. Instead, if k is a field with an absolute value function $|\cdot|: k \rightarrow \mathbf{R}$, then one can approximate $v(T^F u) \approx c$ in the sense that $|v(T^F u) - c|$ is small. Linear and differential cryptanalysis over $\mathbf{F}_2^n$ use the standard absolute value function on the field $k = \mathbf{Q}$ or $\mathbf{R}$. Ultrametric integral cryptanalysis over $\mathbf{F}_2^n$ instead uses the 2-adic absolute value function on $k = \mathbf{Q}$. The p -adic absolute value of an integer x is $|x|_p = p^{-m}$, with p^m the largest power of p dividing x . In particular, $|0|_p = 0$. For a rational number x/y , with x and $y \neq 0$ integers, $|x/y|_p = |x|_p/|y|_p$. In particular, if $v(T^F u)$ is integer-valued and $|v(T^F u) - c|_2 \leq 1/2^\nu$, then $v(T^F u) \equiv c \pmod{2^\nu}$. If $\nu = 1$, this corresponds to an integral property; see Example 1.

One way to construct cryptanalytic methods is to choose bases for the vector spaces $k[X]$ and k^X . This does not affect the properties in Theorem 1. For example, apart from the choice of the absolute value function, linear and (ultrametric) integral cryptanalysis on $\mathbf{F}_2^n$ differ ‘only’ in the choice of basis.

2.2 Multiplicative characters

Linear and integral cryptanalysis are both based on choosing a basis for $k[X]$ (dually, k^X) that diagonalizes the pushforward (dually, pullback) operators of a family of functions [7, 15]. These are the additions $x \mapsto x + r$ for linear cryptanalysis, and the multiplications $x \mapsto r \cdot x$ for integral cryptanalysis. As discussed in [15, Theorem 4.1], functions in k^X that are eigenvectors of the pullback operators of all multiplications are (multiplicative) *characters*.

Definition 1 (Character). *Let k be a field. A character of a monoid X is a homomorphism $\chi: X \rightarrow k$. That is, $\chi(1) = 1$ and $\chi(xy) = \chi(x)\chi(y)$ for all x and y in X .*

The characters of a monoid X form a monoid $\widehat{X}$, where the product of characters χ and ψ is the character $\chi\psi$ defined by $x \mapsto \chi(x)\psi(x)$. The monoid $\widehat{X}$ is called the *dual monoid* of X .

By a result of Dedekind [19, §44], characters are linearly independent. Hence, there exists a basis of k^X consisting of characters if and only if there exist $|X|$ characters. The following theorem provides necessary and sufficient conditions on the ring X for this to hold. A monoid is called *inverse* if, for every element x , there exist a y so that $xyx = x$.

Theorem 2. *Let k be a field and X a finite ring. If X has $|X|$ multiplicative k -valued characters, then X is a commutative inverse monoid with multiplication. If k is algebraically closed and of characteristic zero, then the converse also holds.*

Proof. Let T^r be the pushforward of $x \mapsto r \cdot x$. If there exist $|X|$ characters, then the matrices T^r with r in X are simultaneously diagonalizable. Diagonal matrices commute, so the matrices T^r commute, whence X is commutative. In every finite monoid, there exist positive integers n and $m \geq n$ so that $r^n = r^m$. Since the minimal polynomial of T^r divides $x^{m-n}(x^n - 1)$, the eigenvalues of $(T^r)^n$ are zero or one and $(T^r)^n T^r = T^r$. If $s = r^{n-2}$, then $T^r T^s T^r = T^r$.

It follows that X is an inverse monoid. The converse statement was already mentioned in [15, §4.1] and follows from the representation theory of monoids, see Steinberg [35, Corollary 9.4]. $\square$

The requirement that k is algebraically closed and of characteristic zero is technical; it is necessary to ensure that enough roots of unity exist. Hence, the rings that support a theory of integral cryptanalysis are those that are also commutative inverse multiplicative monoids. Section 2.3 classifies these rings.

2.3 Classification of rings

Theorem 3 shows that a ring that supports a theory of integral cryptanalysis along the same lines as [15] must be a product of fields. The proof assumes some familiarity with commutative algebra.

Theorem 3. *If a finite ring X is a commutative inverse monoid with multiplication, then there exist finite fields $\mathbf{F}_{q_1}, \dots, \mathbf{F}_{q_n}$ such that $X \cong \mathbf{F}_{q_1} \times \mathbf{F}_{q_2} \times \dots \times \mathbf{F}_{q_n}$. More precisely, if S is the set of maximal ideals of X , then $x \mapsto (x + \mathfrak{m} \mid \mathfrak{m} \in S)$ is a ring isomorphism between X and $\prod_{\mathfrak{m} \in S} X/\mathfrak{m}$.*

Proof. Let $\mathfrak{N}$ be the ideal of X containing all elements x such that $x^m = 0$ for some positive integer m . By a well-known theorem of rings [4, Proposition 1.8], $\mathfrak{N}$ is the intersection of all prime ideals of X . Since X is finite, its prime ideals are also maximal ideals. Hence, $\mathfrak{N}$ is the intersection of all maximal ideals. By the Chinese remainder theorem [4, Proposition 1.10], $X/\mathfrak{N} \cong \prod_{\mathfrak{m} \in S} X/\mathfrak{m}$. The product is over the set S of maximal ideals $\mathfrak{m}$ of X . The isomorphism is given by $x \mapsto (x + \mathfrak{m} \mid \mathfrak{m} \in S)$. Since $\mathfrak{m}$ is a maximal ideal, $X/\mathfrak{m}$ is a field.

To complete the proof, we show that $\mathfrak{N} = 0$. Suppose x is a nonzero element of $\mathfrak{N}$, and let $m \geq 2$ be the least integer so that $x^m = 0$. Let y be an element of X so that $xyx = x$, and hence $x^{m-1} = yx^m = 0$. This contradicts the assumption that $m \geq 2$ is the least integer satisfying $x^m = 0$. Hence, $\mathfrak{N} = 0$. $\square$

We make the additional simplification that the characteristic of X is equal to the prime number p used to define the absolute value function. This allows avoiding some technicalities in the choice of the base field k in Section 3.1, but more importantly, the properties of addition are not interesting in a finite field with characteristic different from p . For this reason, from the point of view of applications, nothing is lost by making this simplification. If X has characteristic p , then it is isomorphic to a product of extensions of $\mathbf{F}_p$:

$$X \cong \mathbf{F}_{p^{f_1}} \times \dots \times \mathbf{F}_{p^{f_n}}.$$

Up to the choice of isomorphism, there exists a finite field $\mathbf{F}_q$ that contains all of the fields $\mathbf{F}_{p^{f_1}}, \dots, \mathbf{F}_{p^{f_n}}$ as subfields.

3 Ultrametric integral cryptanalysis

This section generalizes the theory of (ultrametric) integral cryptanalysis from $\mathbf{F}_2^n$ to the rings described in Section 2.3. Although the construction is based on the same principles, and some of the basic properties are analogous, there are fundamental differences. For example, the base field k is not the same, the nature of the character basis is different, and the choice of properties is more subtle.

3.1 Base field

Throughout this paper, a cyclotomic extension of the p -adic numbers $\mathbf{Q}_p$ is used as the base field k . However, no prior familiarity with p -adic numbers is assumed. Before defining k formally, we briefly motivate its choice.

Cryptanalytic properties are combinatorial, so it is reasonable to require that k contains the integers. As discussed in Section 2.2, k should contain enough roots of unity. In particular, it will be shown in Section 3.2 that k must contain a primitive $(q-1)$ -st root of unity ζ , with $q = p^e$ a power of the prime p . The smallest field satisfying these conditions is the number field $\mathbf{Q}(\zeta)$. A common theme in number theory is that it is easier to use *local fields* such as $\mathbf{R}$ and $\mathbf{Q}_p$, obtained by completion with respect to $|\cdot|$ and $|\cdot|_p$ respectively, than *global fields* such as $\mathbf{Q}$ and $\mathbf{Q}(\zeta)$. Ultrametric integral cryptanalysis adheres to this principle, leading to the choice $k = \mathbf{Q}_p(\zeta)$. The ring of integers of $\mathbf{Q}_p(\zeta)$ comes with a natural reduction to $\mathbf{F}_q$. The ‘reduced’ theory yields a version of integral cryptanalysis in characteristic p . For readers unfamiliar with p -adic numbers, we recommend focusing on the modulo- p case on first reading.

The ring of p -adic integers $\mathbf{Z}_p$ consists of all infinite sequences $x = (x_1, x_2, \dots)$ with x_n in $\mathbf{Z}/(p^n)$ so that $x_n \equiv x_{n+1} \pmod{p^n}$ for all $n \geq 1$. Addition and multiplication are defined coordinate-wise, and make $\mathbf{Z}_p$ into an integral domain. The field of p -adic numbers $\mathbf{Q}_p$ is the fraction field of $\mathbf{Z}_p$. As mentioned above, the field k is the extension $\mathbf{Q}_p(\zeta)$ of $\mathbf{Q}_p$, where ζ is a primitive $(q-1)$ -st root of unity. The ring of integers of $\mathbf{Q}_p(\zeta)$ is the integral domain $\mathbf{Z}_p[\zeta]$, which has a unique maximal ideal (p) . The p -adic valuation $\text{ord}_p x$ of $x \neq 0$ in $\mathbf{Z}_p[\zeta]$ is the largest power of p that divides x . The p -adic absolute value of x is $|x|_p = p^{-\text{ord}_p x}$. In addition, we define $\text{ord}_p 0 = \infty$ and $|0|_p = 0$. The p -adic valuation of x/y in $\mathbf{Q}_p(\zeta)$ is $\text{ord}_p x - \text{ord}_p y$, and the absolute value is $|x|_p/|y|_p$. It satisfies the ultrametric triangle inequality $|a+b|_p \leq \max\{|a|_p, |b|_p\}$. Equivalently, $\text{ord}_p(a+b) \geq \min\{\text{ord}_p a, \text{ord}_p b\}$. For further background on p -adic numbers, see [29].

Example 2. If $q = p$, then $\mathbf{Q}_p$ contains a complete set of $(q-1)$ -st roots of unity so that $\mathbf{Q}_p(\zeta) = \mathbf{Q}_p$. For example, $\mathbf{Z}_5$ contains a fourth root of unity $x = (x_1, x_2, \dots)$ with $x_1 \equiv 2 \pmod{5}$. The digits $x_2, x_3, \dots$ can be found as follows. Since $x_2 = x_1 + 5t$, we must have $(2+5t)^4 \equiv 1 \pmod{25}$. That is, $t \equiv 1 \pmod{5}$ (uniquely modulo five) and $x_2 = 7$. Repeating this process ad infinitum is called Hensel lifting [29, Theorem 3]. The 5-adic number $x = (2, 7, 57, \dots)$ satisfies $x^4 = 1$. Another way to write x is as the series $x = 2 + 1 \cdot 5 + 2 \cdot 5^2 + 1 \cdot 5^3 + 3 \cdot 5^4 + \dots$ that converges with respect to the 5-adic absolute value.

The quotient ring $\mathbf{Z}_p[\zeta]/(p)$ is a finite field $\mathbf{F}_q$, namely the extension field $\mathbf{F}_p(\zeta)$ defined by the same irreducible polynomial as $\mathbf{Q}_p(\zeta)$. It is standard to call $\mathbf{F}_q$ the *residue field* of $\mathbf{Q}_p(\zeta)$. Elements of the integer ring of $\mathbf{Q}_p(\zeta)$ can be reduced modulo p to obtain elements of $\mathbf{F}_q$. There is also a natural way to map elements of $\mathbf{F}_q$ to $\mathbf{Q}_p(\zeta)$, see [29, §I.4, Exercise 13].

Lemma 1. *For every $x \neq 0$ in the residue field $\mathbf{F}_q$ of $\mathbf{Q}_p(\zeta)$, there exists a unique $(q-1)$ -st root of unity $\tau(x)$ in $\mathbf{Z}_p[\zeta]$ so that $\tau(x) \equiv x \pmod{p}$. Furthermore, the map $\tau: \mathbf{F}_q \rightarrow \mathbf{Q}_p(\zeta)$ defined by $x \mapsto \tau(x)$ and $\tau(0) = 0$ is a character of $\mathbf{F}_q$.*

The character τ in Lemma 1 is called the Teichmüller character. It is the main ingredient in the construction of the character basis in Section 3.2.

Example 3. Let $\tau: \mathbf{F}_5 \rightarrow \mathbf{Q}_5$ be the Teichmüller character for $q = p = 5$. By definition, $\tau(0) = 0$. The values $\tau(1) = 1$ and $\tau(4) = -1$ follow from the fact that $\mathbf{Q}$ contains the roots of unity ± 1 . In Example 2, $\tau(2)$ was computed as $\tau(2) = 2 + 1 \cdot 5 + \dots$. The value of $\tau(3)$ follows from $\tau(3) = \tau(4)\tau(2) = -\tau(2)$.

3.2 Multiplicative characters

The construction of the bases for $k[X]$ and k^X relies on the characters of a ring X isomorphic to a product of fields $\mathbf{F}_{q_1} \times \dots \times \mathbf{F}_{q_n}$. As discussed in Section 2.3, it can be assumed that all these fields are contained in a single field $\mathbf{F}_q$. If $\chi_1, \dots, \chi_n$ are multiplicative characters of $\mathbf{F}_{q_1}, \dots, \mathbf{F}_{q_n}$ respectively, then the function $\chi = \chi_1 \otimes \dots \otimes \chi_n$ defined by

$$\chi((x_1, \dots, x_n)) = \prod_{i=1}^n \chi_i(x_i),$$

is a character of the product ring $\mathbf{F}_{q_1} \times \dots \times \mathbf{F}_{q_n}$. To obtain the characters of X , it is sufficient to pull χ back along the isomorphism $f: X \rightarrow \mathbf{F}_{q_1} \times \dots \times \mathbf{F}_{q_n}$. That is, $\chi \circ f$ is a character of X .

By the above argument, it is enough to determine the characters of the fields $\mathbf{F}_{q_i}$. For every exponent v_i in $\{0, 1, \dots, q_i - 1\}$, there is a multiplicative homomorphism $\chi_i: \mathbf{F}_{q_i} \rightarrow k$ given by $\chi_i(x) = \tau(x^{v_i})$, with τ the Teichmüller character (see Lemma 1 in Section 3.1). Explicitly, the corresponding character $\chi = \chi_1 \otimes \dots \otimes \chi_n$ is given by $\chi(x) = \prod_{i=1}^n \tau(x_i^{v_i})$ for $x = (x_1, \dots, x_n)$. Observe that in the residue field $\mathbf{F}_q$ of k , i.e. after reduction modulo p , the characters are monomial functions $x \mapsto x^v$.

3.3 Basis and dual basis

As discussed in Section 2.2, the characters of X form a basis of k^X . To emphasize that we think of k^X as dual to $k[X]$, we denote these basis vectors by b^χ with χ a character. That is, $b^\chi(\delta_x) = \chi(x)$. Following Section 2.1, there exists a basis

for $k[X]$ dual to the basis of characters. Denote the corresponding basis vectors by b_ψ , with ψ a character, so that

$$b^\chi(b_\psi) = \begin{cases} 1 & \text{if } \chi = \psi, \\ 0 & \text{else.} \end{cases} \quad (1)$$

To determine b_ψ explicitly, it is again sufficient to do this for individual fields $\mathbf{F}_{q_i}$. Indeed, if $\psi = \psi_1 \otimes \cdots \otimes \psi_n$, then $b_\psi = b_{\psi_1} \otimes \cdots \otimes b_{\psi_n}$ by a general property of dual bases. If $\psi_i = 1$, then $b_{\psi_i} = \delta_0$. Otherwise,

$$b_{\psi_i} = \frac{1}{q_i - 1} \sum_{x \in \mathbf{F}_{q_i}^\times} \psi_i(1/x) \delta_x + \begin{cases} -\delta_0 & \text{if } \psi_i(x) = \tau(x^{q_i-1}) \text{ for all } x \in \mathbf{F}_{q_i}, \\ 0 & \text{else.} \end{cases} \quad (2)$$

To prove that this is correct, it is enough to verify (1). The verification for $\psi_i = 1$ is immediate. For the remaining cases, it follows from the orthogonality of group characters: $\sum_{x \in \mathbf{F}_{q_i}^\times} \psi_i(1/x) \chi_i(x) = q_i - 1$ if $\psi_i = \chi_i$ and zero otherwise.

Definition 2. Let X be a finite ring of characteristic p isomorphic to a product of fields, with multiplicative characters $\widehat{X}$. The ultrametric integral change-of-basis is the linear map $\mathcal{U}_X: k[X] \rightarrow k[\widehat{X}]$ defined by $\mathcal{U}_X(b_\chi) = \delta_\chi$ for all multiplicative characters χ .

If $X = \mathbf{F}_2^n$, then Definition 2 coincides with [15, Definition 4.2]. The modulo-two reduction of this case was discussed earlier in [14], and implicitly in [16]. By construction of the basis functions, for a product of fields,

$$\mathcal{U}_{\mathbf{F}_{q_1} \times \cdots \times \mathbf{F}_{q_n}} = \mathcal{U}_{\mathbf{F}_{q_1}} \otimes \cdots \otimes \mathcal{U}_{\mathbf{F}_{q_n}}.$$

An important difference with [15] is that the coordinates of $\mathcal{U}_X$ are no longer integers, but elements of $\mathbf{Z}_p[\zeta]$. The reason for this is that, for $p = 2$, the required root of unity (i.e. 1) is an integer.

Example 4. Let $X = \mathbf{F}_5$. The rows of $\mathcal{U}_X$ correspond to the characters $x \mapsto \tau(x^v)$, where $\tau(0) = 0$, $\tau(1) = 1$, $\tau(2) = i$, $\tau(3) = -i$ and $\tau(4) = -1$. Here, $i = 2 + 1 \cdot 5 + \cdots$ is the unique 5-adic square root of -1 so that $i \equiv 2 \pmod{5}$.

$$\mathcal{U}_X = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & i & -i & -1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 1 & -i & i & -1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad \text{and} \quad \mathcal{U}_X^{-1} = \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 & 0 & -4 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & -i & -1 & i & 1 \\ 0 & i & -1 & -i & 1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix}.$$

The columns of $\mathcal{U}_X^{-1}$ are the basis vectors b_ψ . Note that all coordinates of $\mathcal{U}_X$ and $\mathcal{U}_X^{-1}$ are integer elements of $k = \mathbf{Q}_5$. This is true in general.

3.4 Properties

As explained in Section 2.1, cryptanalytic properties of a function $F: X \rightarrow Y$ are pairs (u, v) , with u a vector in $k[X]$ and v a function in k^Y . Let $\hat{u} = \mathcal{U}_X(u)$ and $\hat{v} = \mathcal{U}_Y^{-\vee}(v)$ with $\mathcal{U}_Y^{-\vee}$ the dual change-of-basis transformation of $\mathcal{U}_Y$, i.e. $\mathcal{U}_Y^{-\vee}(b^\chi) = \delta^\chi$ for all multiplicative characters χ . The property (u, v) is evaluated as follows:

$$v(T^F u) = \hat{v}(\mathcal{U}_Y T^F \mathcal{U}_X^{-1} \hat{u}).$$

If $u = b_\psi$ and $v = b^\chi$, then the result is a coordinate of the matrix representation of $\mathcal{U}_Y T^F \mathcal{U}_X^{-1}$. Section 3.5 discusses the properties of this matrix in detail. Ultrametric integral cryptanalysis reduces the evaluation of all cryptanalytic properties to the analysis of these matrices.

To reduce data-complexity, one can choose u from the span $\langle \delta_x \mid x \in S \rangle$ with S a subset of X . For example, $u = \delta_S = \sum_{x \in S} \delta_x$. Ideally, the set S is as small as possible. The multiplicative submonoids of X are a particularly well-behaved choice of S . This is analogous to multidimensional linear cryptanalysis, see for example [8, Theorem 4.1] or [9, Example 3.10], where it is the additive subgroups that play the role of the submonoids.

Up to isomorphism, X is a product of fields, i.e. $\mathbf{F}_{q_1} \times \cdots \times \mathbf{F}_{q_n}$. Hence, the multiplicative submonoids of X are (isomorphic to) products of the multiplicative submonoids of the fields $\mathbf{F}_{q_1}, \dots, \mathbf{F}_{q_n}$. Every submonoid of $\mathbf{F}_{q_i}$ is either a multiplicative subgroup of $\mathbf{F}_{q_i}^\times$, or of the form $\{0\} \cup G$ with G a multiplicative subgroup. Since $\mathcal{U}_{\mathbf{F}_q}(\delta_0) = \delta_1$, one has (with $\oplus$ the direct sum)

$$\mathcal{U}_{\mathbf{F}_q}(\langle \delta_x \mid x \in \{0\} \cup G \rangle) = \mathcal{U}_{\mathbf{F}_q}(\langle \delta_x \mid x \in G \rangle) \oplus \langle \delta_1 \rangle.$$

In particular, $\mathcal{U}_{\mathbf{F}_q}(\delta_{\{0\} \cup G}) = \delta_1 + \mathcal{U}_{\mathbf{F}_q}(\delta_G)$. For subgroups of $\mathbf{F}_q^\times$, the following theorem provides the result.

Theorem 4. *Let G be a subgroup of $\mathbf{F}_q^\times$ with index $n = (q-1)/|G|$. If H is the monoid of characters χ with $\chi^n = 1$, then $\mathcal{U}_{\mathbf{F}_q}(\langle \delta_x \mid x \in G \rangle) = \langle \delta_{\chi H} \mid \chi \in \widehat{\mathbf{F}}_q \rangle$. In particular, the indicator δ_G of G satisfies $\mathcal{U}_{\mathbf{F}_q}(\delta_G) = |G| \delta_H$.*

Proof. Let g be a generator of the cyclic group $\mathbf{F}_q^\times$. The subgroup G is cyclic with generator g^n , where n is index of G . Let $\chi\psi_1$ and $\chi\psi_2$ be characters in χH . Since $\psi_1(g^{ni}) = 1$ and $\psi_2(g^{ni}) = 1$ for all integers i , it holds that $\chi\psi_1|_G = \chi\psi_2|_G$. Equivalently, $b^{\chi\psi_1}(\delta_x) = b^{\chi\psi_2}(\delta_x)$ for all x in G , so elements of $\mathcal{U}_{\mathbf{F}_q}(\langle \delta_x \mid x \in G \rangle)$ are constant on χH . This shows that $\mathcal{U}_{\mathbf{F}_q}(\langle \delta_x \mid x \in G \rangle) \subseteq \langle \delta_{\chi H} \mid \chi \in \widehat{\mathbf{F}}_q \rangle$. To prove the equality, note that both spaces have dimension $|G| = (q-1)/|H|$. Finally, for $\mathcal{U}_{\mathbf{F}_q}(\delta_G)$, note that

$$b^\chi(\delta_G) = \sum_{x \in G} \chi(x) = \sum_{i=0}^{|G|-1} \chi(g)^{ni} = \begin{cases} |G| & \text{if } \chi^n = 1, \\ 0 & \text{else.} \end{cases}$$

The case $\chi^n \neq 1$ follows from the orthogonality of group characters. Hence, $\mathcal{U}_{\mathbf{F}_q}(\delta_G)$ is equal to $|G| \delta_H$. $\square$

To use a submonoid of the form $\{0\} \cup G$, the choice $u = \delta_G - |G| \delta_0$ is better than $\delta_{\{0\} \cup G}$. The data-complexity is the same, but the support of $\hat{u}$ does not include the trivial character (higher exponents lead to stronger properties). An example is given in Section A.1.

By taking tensor products, Theorem 4 is sufficient to deal with all submonoids. The applications in Sections 6 and 7 use submonoids corresponding to an ideal $(v) = v\mathbf{F}_q^n$ of $\mathbf{F}_q^n$ generated by v in $\{0, 1\}^n$. These are sets of tuples where the coordinates with $v_i = 0$ are fixed to zero and the other coordinates are free. By Theorem 4, $\delta_{\mathbf{F}_q} = qb_1 + (q-1)b_\chi$ with $\chi(x) = \tau(x^{q-1})$ for all x . Hence,

$$\delta_{(v)} = \sum_{\psi} q^{n-\text{wt } \psi} (q-1)^{\text{wt } \psi} b_{\psi},$$

where the sum is over all characters ψ of the form $\psi(x) = \tau(x^u)$ with u in $\{0, 1\}^n$ and $u \preceq v$ (coordinate-wise inequality), and $\text{wt } \psi$ is the Hamming weight of u .

3.5 Propagation

As mentioned at the beginning of Section 3.4, the evaluation of properties in ultrametric integral cryptanalysis is based on the transition matrix $\mathcal{U}_Y T^F \mathcal{U}_X^{-1}$. These matrices share all the properties of T^F discussed in Section 2.1.

Definition 3. For a function $F: X \rightarrow Y$ between finite rings X and Y of characteristic p isomorphic to a product of fields, let $A^F = \mathcal{U}_Y T^F \mathcal{U}_X^{-1}$. The ultrametric integral transition matrix of F is the matrix representation of A^F with respect to the standard bases of $k[\hat{X}]$ and $k[\hat{Y}]$ respectively.

For $F: \mathbf{F}_2^n \rightarrow \mathbf{F}_2^m$, Definition 3 coincides with [15, Definition 4.2]. In this case, it was shown in [15] that the matrix A^F has integer coordinates. This is no longer true, but it is still true that the coordinates of A^F are integral elements of k . Hence, they can be reduced modulo p , and like in [15] this yields the coefficients of the algebraic normal form of products of the coordinates of F . The algebraic normal form of a function $\mathbf{F}_{q_1} \times \cdots \times \mathbf{F}_{q_n} \rightarrow \mathbf{F}_q$ with $\mathbf{F}_{q_1}, \dots, \mathbf{F}_{q_n}$ subfields of $\mathbf{F}_q$, is its unique interpolating polynomial in $\mathbf{F}_q[x_1, \dots, x_n]/(x_1^{q_1} - x_1, \dots, x_n^{q_n} - x_n)$.

Theorem 5. Let $F: \prod_{i=1}^n \mathbf{F}_{p^{e_i}} \rightarrow \prod_{i=1}^m \mathbf{F}_{p^{f_i}}$ be a function with ultrametric integral transition matrix A^F . For all characters ψ and χ with $\psi(x) = \tau(x^u)$ and $\chi(x) = \tau(x^v)$, the coordinate $A_{\chi, \psi}^F$ is an integral element of k congruent modulo p to the coefficient of x^u in the algebraic normal form of F^v .

Proof. An element of k is integral if and only if its p -adic absolute value is less than one. The coordinates of A^F are integral because $A_{\chi, \psi}^F = b^\chi(T^F b_\psi)$ and the coordinates of b_ψ and b^χ are themselves integral. In particular, since $\chi(x)$ is either zero or a root of unity, $|\chi(x)|_p \leq 1$ for all x . For the coordinates of b_ψ , this follows from (2) by the same argument; note that $|p^{f_i} - 1|_p = 1$.

The coordinates of A^F are integral elements of k , so their reductions modulo p are well-defined elements of $\mathbf{F}_q$. By definition, the transpose of T^F satisfies

$T^{F^\vee} b^\chi = \sum_{\psi} A_{\chi, \psi}^F b^\psi$. Since $b^\psi(\delta_x) \equiv x^u \pmod{p}$ and $b^\chi(\delta_x) \equiv x^v \pmod{p}$, reduction modulo p shows that

$$F^v(x) \equiv \sum_{\psi: x \mapsto \tau(x^u)} A_{\chi, \psi}^F x^u \pmod{p},$$

for all x in $\prod_{i=1}^n \mathbf{F}_{p^{e_i}}$. The algebraic normal form of F^v is the *unique* interpolating polynomial F^v of degree less than $p^{e_i} - 1$ in x_i , so the coordinates $A_{\chi, \psi}^F \pmod{p}$ are equal to the coefficients of the algebraic normal form of F^v . $\square$

A few examples of Definition 3 and Theorem 5 are listed in Section A.2. Ultrametric integral transition matrices satisfy all of the properties listed in Theorem 1. In particular, if $F = F_r \circ \dots \circ F_2 \circ F_1$, then $A^F = A^{F_r} \dots A^{F_2} A^{F_1}$. In terms of coordinates, if χ_1 and χ_{r+1} are characters, then

$$A_{\chi_{r+1}, \chi_1}^F = \sum_{\chi_2, \dots, \chi_r} \prod_{i=1}^r A_{\chi_{i+1}, \chi_i}^{F_i}, \quad (3)$$

where the sum is over all intermediate characters $\chi_2, \dots, \chi_r$. Similar to ultrametric integral cryptanalysis over $\mathbf{F}_2^n$, a sequence $(\chi_1, \dots, \chi_{r+1})$ will be called a trail. The product $\prod_{i=1}^r A_{\chi_{i+1}, \chi_i}^{F_i}$ is its correlation. In practice, only a set of dominant trails with large absolute correlation will be taken into account. This is discussed in Section 5.1.

To compute the correlations of trails, the coordinates of the matrices A^{F_i} need to be known for the basic operations used to construct primitives. Theorem 6 provides this for operations compatible with the multiplicative structure of the ring: multiplication by a constant, and homomorphisms of the underlying multiplicative monoid.

Theorem 6. *The ultrametric integral transition matrices of constant multiplication and monoid homomorphisms are given by*

(1) *If $F: X \rightarrow X$ is defined by $F(x) = m \cdot x$, then*

$$A_{\chi, \psi}^F = \begin{cases} \chi(m) & \text{if } \psi = \chi, \\ 0 & \text{else.} \end{cases}$$

(2) *If $F: X \rightarrow Y$ is a homomorphism of multiplicative monoids, then*

$$A_{\chi, \psi}^F = \begin{cases} 1 & \text{if } \psi = \chi \circ F, \\ 0 & \text{else.} \end{cases}$$

Proof. If F is a multiplication, then by definition b^χ is an eigenvector of $A^{F^\vee}$. Hence, $A_{\chi, \psi}^F = b^\chi(T^F b_\psi) = \chi(m) b^\chi(b_\psi)$. The result follows by the definition of dual bases. If F is a homomorphism of multiplicative monoids, then $A_{\chi, \psi}^F = b^\chi(T^F b_\psi) = b^{\chi \circ F}(b_\psi)$ with $\chi \circ F$ a well-defined character of X . Again, the result follows by the definition of dual bases. $\square$

Copy and multiplication operations are two important examples of monoid homomorphisms. Theorem 6 implies the propagation rules illustrated in Figure 1. Another important class of homomorphisms, used in all of the applications in Section 7, are power functions. If $F: X \rightarrow X$ is given by $F(x) = x^d$, then Theorem 6 shows that $A_{\chi, \psi}^F = 1$ if $\psi = \chi^d$ and zero otherwise.



Fig. 1: Propagation rules for copy and multiplication operations.

4 Addition

Multiplication and other monoid homomorphisms are particularly well-behaved in integral cryptanalysis, but addition is not. This section describes the propagation rules for addition with a constant, and for addition between two variables.

4.1 Weight of characters

For an addition operation F , the p -adic valuation of $A_{\chi, \psi}^F$ admits a simple expression depending on the *weight* of the characters ψ and χ . The weight of a character is closely related to the degree of the corresponding monomial obtained by reduction modulo p .

Let q be a prime power and r an integer root of q . That is, if $q = p^e$, then $r = p^i$ with i a divisor of e . The r -weight of a character $\chi: \mathbf{F}_q \rightarrow k$ given by $\chi(x) = \tau(x^v)$ with exponent v in $\{0, 1, \dots, q-1\}$ is the sum of the base- r digits of v . This will be denoted as $\text{wt}_r \chi = s_r(v)$. In particular, $\text{wt}_r \chi$ is the degree of the algebraic normal form of the monomial function $x \mapsto \chi(x) \equiv x^v \pmod{p}$ as a function on $\mathbf{F}_r^l$ with $l = e/i$. For a character $\chi = \chi_1 \otimes \dots \otimes \chi_n$ of a product of fields, the weight is defined additively: $\text{wt}_r \chi = \sum_{i=1}^n \text{wt}_r \chi_i$. Throughout this section, the value of r is always in $\{p, q\}$. In Section 6, other values are used as well. The weight of characters satisfies the inequality $\text{wt}_r \chi\psi \leq \text{wt}_r \chi + \text{wt}_r \psi$. This follows from Lemma 2. This result is known, going back to [30, page 116].

Lemma 2. *Let r be a positive integer. For all non-negative integers x and y , $s_r(x+y) = s_r(x) + s_r(y) + (r-1)c(x, y)$, with $c(x, y)$ the number of carries when x and y are added in base r . In particular, $s_r(x+y) \leq s_r(x) + s_r(y)$.*

Lemma 2 has the following corollary that plays a technical role in Section 4.2.

Lemma 3. *Let r be a positive integer and q a power of r . For all non-negative integers $x < q$ and k , $s_r(x) \leq s_r(x + k(q - 1))$.*

Proof. Lemma 2 implies $s_r(x - y) \geq s_r(x) - s_r(y)$ for all non-negative integers x and $y \leq x$. Indeed, since $x = (x - y) + y$, this follows from the inequality stated at the end of Lemma 2. In particular, $s_r(x + k(q - 1))$ is bounded from below by

$$s_r(x + k(q - 1)) \geq s_r(x + kq) - s_r(k) = s_r(x) + s_r(kq) - s_r(k) = s_r(x).$$

The first equality is due to the fact that q is a power of r and $x < q$, so that the base- r digits of $x + kq$ are the base- r digits of x followed by those of k . Finally, $s_r(kq) = s_r(k)$, because q is a power of the base r . $\square$

4.2 Addition with a constant

In Section 4.3, it is shown that the transition matrix of an addition between two operands can be deduced from that of addition with a constant. Hence, this section first discusses addition with a constant.

Before turning to the general case, it is instructive to consider the simpler ‘modulo p ’ variant. By Theorem 5, the coordinates of A^F modulo p are the coefficients of the algebraic normal form of F^v . For example, if $F: \mathbf{F}_q \rightarrow \mathbf{F}_q$ is the function defined by $F(x) = x + 1$, then the algebraic normal form of F^v is equal to

$$(x + 1)^v = \sum_{i=0}^v \binom{v}{i} x^i$$

Hence, if $\psi(x) = \tau(x^u)$ and $\chi(x) = \tau(x^v)$, then $A_{\chi, \psi}^F \equiv \binom{v}{u} \pmod{p}$. For the applications in this paper, it is enough to know whether or not $A_{\chi, \psi}^F \equiv 0 \pmod{p}$. The following theorem of Kummer [30, page 116] shows that this is the case if and only if $s_p(v) + s_p(v - u) > s_p(u)$.

Theorem 7 (Kummer). *For all non-negative integers n and m , $\text{ord}_p \binom{n}{m}$ is equal to the number of carries when adding m to $n - m$ in base p . That is,*

$$\text{ord}_p \binom{n}{m} = \frac{s_p(m) + s_p(n - m) - s_p(n)}{p - 1}.$$

The above strategy does not allow to compute the divisibility of the coordinates of A^F by higher powers of p . However, the idea can be extended using the following lemma. This result is known, see for example [34, §4, Lemma 1]

Lemma 4. *Let x and y be integral elements of k , and n a positive integer. If $x \equiv y \pmod{p^n}$, then $x^p \equiv y^p \pmod{p^{n+1}}$.*

Proof. If $x \equiv y \pmod{p^n}$ then $y = x + mp^n$ for some integral element m of k . By the binomial formula, $y^p = (x + mp^n)^p = \sum_{i=0}^p \binom{p}{i} x^{p-i} (mp^n)^i$. All terms on the right-hand side are divisible by p^{n+1} , except for $i = 0$. Since this term equals x^p , the result follows. $\square$

Theorem 8 follows from the combination of Lemma 1 and Theorem 7. This result extends to addition by a constant in all finite rings of characteristic p that are isomorphic to a product of fields. In particular, observe that $F(x) = x + t = (x_1 + t_1, \dots, x_n + t_n)$. Hence, $A^F = A^{F_1} \otimes \dots \otimes A^{F_n}$ with $F_i(x) = x + t_i$, and $\text{ord}_p A_{\chi, \psi}^F = \sum_{i=1}^n \text{ord}_p A_{\chi_i, \psi_i}^{F_i}$ for characters $\psi = \psi_1 \otimes \dots \otimes \psi_n$ and $\chi = \chi_1 \otimes \dots \otimes \chi_n$. Since the expression for the valuation in Theorem 8 is linear in the weights of the characters, it carries over to arbitrary products of fields.

Theorem 8. *For t in $\mathbf{F}_q^\times$, let $F: \mathbf{F}_q \rightarrow \mathbf{F}_q$ be defined by $F(x) = x + t$. For all characters ψ and $\chi \neq 1$ of the multiplicative monoid $\mathbf{F}_q$,*

$$\text{ord}_p A_{\chi, \psi}^F \geq \frac{\text{wt}_p \psi + \text{wt}_p \rho - \text{wt}_p \chi}{p-1}.$$

Here, ρ is the multiplicative character defined by $\rho(x) = \tau(x^{v-u})$ if $u \leq v$ and $\rho(x) = \tau(x^{q-1+v-u})$ otherwise, for $\psi(x) = \tau(x^u)$ and $\chi(x) = \tau(x^v)$.

Proof. It suffices to prove the result for $t = 1$, since $F(x) = t(x/t + 1)$. Indeed, by Theorem 6, the ultrametric integral transition matrices of multiplication and division by t are diagonal. Hence, the correlation for $t = 1$ is the same up to a factor $\chi(t)/\psi(t)$ with valuation zero. Since χ is a character, $\chi(x+1)^q = \chi(x+1)$ for all x in $\mathbf{F}_q$. Furthermore, if $\chi(x) = \tau(x)^v$, then $\chi(x+1) \equiv (x+1)^v \equiv (\tau(x)+1)^v \pmod{p}$, so Lemma 4 implies that

$$\chi(x+1) \equiv (\tau(x)+1)^{qv} \equiv \sum_{i=0}^{qv} \binom{qv}{i} \tau(x)^i \pmod{pq}.$$

Let $\omega_i: \mathbf{F}_q \rightarrow k$ be the character defined by $\omega_i(x) = \tau(x)^i$. In terms of functions, the above equality then becomes

$$T^{F^\vee} b^\chi \equiv \sum_{i=0}^{qv} \binom{qv}{i} b^{\omega_i} \pmod{pq}.$$

Since $A_{\chi, \psi}^F = (T^{F^\vee} b^\chi)(b_\psi)$, it is sufficient to evaluate the above expression in b_ψ . The only terms that remain are those with $\omega_i = \psi$. If $\psi = 1$, then the sum contains only one term ($i = 0$) and the result follows. For the remainder of the proof, choose u and w in $\{1, \dots, q-1\}$ so that $\psi = \omega_u$ and $\rho = \omega_w$. By Theorem 7,

$$\text{ord}_p \binom{qv}{i} = \frac{s_p(i) + s_p(qv-i) - s_p(qv)}{p-1} = \frac{s_p(i) + s_p(qv-i) - s_p(v)}{p-1}.$$

There exists a non-negative integer n so that $i = u + n(q-1)$. Hence, by Lemma 3, $s_p(i) = s_p(u + n(q-1)) \geq s_p(u)$. Similarly, there exists a non-negative integer m so that $qv-i = w + m(q-1)$. Again, by Lemma 3, $s_p(qv-i) = s_p(w + m(q-1)) \geq s_p(w)$. It follows that

$$\text{ord}_p \binom{qv}{i} \geq \frac{s_p(u) + s_p(w) - s_p(v)}{p-1} = \frac{\text{wt}_p \psi + \text{wt}_p \rho - \text{wt}_p \chi}{p-1}.$$

Since this holds for every term, the result follows. $\square$

The inequality in Theorem 8 turns out to be an equality. More precisely, one has the following closed-form formula for the coordinates of A^F in terms of the p -adic gamma function Γ_p :

$$A_{\chi,\psi}^F = (-1)^e \frac{(-p)^\nu}{1-q} \prod_{i=0}^{e-1} \frac{\Gamma_p\left(\frac{\text{wt}_q \chi^{p^i}}{1-q} + 1\right)}{\Gamma_p\left(\frac{\text{wt}_q \psi^{p^i}}{1-q} + 1\right) \Gamma_p\left(\frac{\text{wt}_q \rho^{p^i}}{1-q} + 1\right)}, \quad (4)$$

with ν the lower bound for the valuation from Theorem 8. Since this result is not used in this paper and the proof requires additional background, it is omitted.

Example 5. If $F: \mathbf{F}_5 \rightarrow \mathbf{F}_5$ is the addition $F(x) = x + 1$, then

$$A^F = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & -\frac{1}{4} & \beta & \beta & -\frac{5}{4} \\ 1 & \alpha - \frac{1}{4} & -\beta & -\frac{5}{4} & -\frac{5}{4} \\ 1 & -\alpha & -\alpha & -\frac{1}{4} & -\frac{5}{4} \\ 1 & \frac{1}{4} & -\frac{1}{4} & \frac{1}{4} & -\frac{1}{4} \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 4 & 0 & 0 & 0 \\ 1 & 3 & 4 & 0 & 0 \\ 1 & 2 & 2 & 4 & 0 \\ 1 & 1 & 4 & 1 & 4 \end{bmatrix} \pmod{5},$$

with $\alpha = (2i - 1)/4$ and $\beta = (2i + 1)/4$, for $i = \tau(2)$. The value $\beta = 2 \cdot 5 + 2 \cdot 5^2 + 4 \cdot 5^3 + \dots$ satisfies $\text{ord}_5 \beta = 1$.

The properties of addition are more interesting for extension fields, as illustrated by the following example for $q = 4$.

Example 6. Let $F: \mathbf{F}_4 \rightarrow \mathbf{F}_4$ be the addition $F(x) = x + 1$ with $\mathbf{F}_4 = \mathbf{F}_2(\alpha)$ so that $\alpha^2 + \alpha + 1 = 0$. In this case, A^F is given by

$$A^F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & -\frac{1}{3} & \frac{2}{3} & -\frac{4}{3} \\ 1 & \frac{2}{3} & -\frac{1}{3} & -\frac{4}{3} \\ 1 & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{4} \end{bmatrix} \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \pmod{2},$$

where the rows are indexed by field elements $0, 1, \alpha$ and $\alpha + 1$ (in that order).

4.3 Addition of two variables

An addition between two variables is a function $F: X \times X \rightarrow X$ defined by $F(x, y) = x + y$. Like for additions with a constant, it is enough to consider the case of finite fields. The result for arbitrary products of fields follows by taking tensor products.

Theorem 9. *Let $F: \mathbf{F}_q \times \mathbf{F}_q \rightarrow \mathbf{F}_q$ be the function defined by $F(x, y) = x + y$. For all characters χ of $\mathbf{F}_q$ and $\psi = \psi_1 \oplus \psi_2$ of $\mathbf{F}_q \times \mathbf{F}_q$, we have $A_{\chi,\psi}^F = 0$ unless $\chi = \psi_1 \psi_2$. If $\chi \neq 1$, then*

$$\text{ord}_p A_{\chi,\psi}^F \geq \frac{\text{wt}_p \psi - \text{wt}_p \chi}{p - 1}.$$

Proof. Let $F_t: \mathbf{F}_q \rightarrow \mathbf{F}_q$ be the function defined by $F_t(x) = x + t$. As mentioned in the proof of Theorem 8, $A_{\chi, \psi}^{F_t} = \chi(t)/\psi(t)A_{\chi, \psi}^{F_1}$. This makes it possible to relate the coordinates of A^F to those of A^{F_1} :

$$A_{\chi, \psi}^F = \frac{1}{q-1} \sum_{t \in \mathbf{F}_q^\times} \frac{A_{\chi, \psi_1}^{F_t}}{\psi_2(t)} = \frac{A_{\chi, \psi_1}^{F_1}}{q-1} \sum_{t \in \mathbf{F}_q^\times} \frac{\chi(t)}{(\psi_1 \psi_2)(t)} = \begin{cases} A_{\chi, \psi_1}^{F_1} & \text{if } \chi = \psi_1 \psi_2, \\ 0 & \text{else.} \end{cases}$$

The last step relies on the orthogonality of the characters of the multiplicative group $\mathbf{F}_q^\times$. The result now follows from Theorem 8, and by writing $\text{wt}_p \psi_1 + \text{wt}_p \psi_2 = \text{wt}_p \psi$. $\square$

Like for Theorem 8, the lower bound in Theorem 9 is actually tight. In fact, based on the proof of Theorem 8, a closed-form formula for the coordinates of A^F similar to (4) can be given. For the applications in Sections 6 and 7, the valuation lower bound from Theorem 9 is sufficient.

Example 7. Let $F: \mathbf{F}_4 \times \mathbf{F}_4 \rightarrow \mathbf{F}_4$ be the addition function $F(x, y) = x + y$ of the field $\mathbf{F}_4 = \mathbf{F}_2(\alpha)$ with $\alpha^2 + \alpha + 1 = 0$. A short calculation shows that

$$A^F = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & -\frac{4}{3} & 0 & 0 & \frac{2}{3} & 0 & 0 & -\frac{4}{3} & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{2}{3} & 0 & 0 & 1 & 0 & 0 & -\frac{4}{3} & 0 & 0 & -\frac{4}{3} \\ 0 & 0 & 0 & 1 & 0 & 0 & -\frac{1}{3} & 0 & 0 & -\frac{1}{3} & 0 & 0 & 1 & 0 & 0 & -\frac{4}{3} \end{bmatrix}.$$

Compare this to the transition matrix for addition with a constant in Example 6.

The following equivalent formulation of Theorem 9 is used in the applications:

$$A_{\chi, \psi}^F \geq \sum_{i=0}^{e-1} \frac{\text{wt}_q \psi^{p^i} - \text{wt}_q \chi^{p^i}}{q-1}. \quad (5)$$

This follows from the fact that $(\text{wt}_p \psi - \text{wt}_p \chi)/(p-1)$ is equal to the number of carries when the exponents of ψ_1 and ψ_2 are added in base p . In the above formula, $(\text{wt}_q \psi^{p^i} - \text{wt}_q \chi^{p^i})/(q-1)$ is one if a carry occurs at the highest digit in the addition of the exponents of $\psi_1^{p^i}$ and $\psi_2^{p^i}$, and zero otherwise. Since the base- p digits of $\text{wt}_q \chi^{p^i}$ are a rotation of the digits of $\text{wt}_q \chi$ over i positions, summing over all i in $\{0, 1, \dots, e-1\}$ yields the total number of carries.

5 Practical trail bounding methods

Sections 3.4 and 3.5 show that, like for ultrametric integral cryptanalysis over $\mathbf{F}_2^n$, properties can be evaluated by summing the correlations of trails. In practice, enumerating all trails is rarely feasible, so we introduce two methods to avoid this. The first method is the principle of dominant trails. This is mostly the same as over $\mathbf{F}_2^n$, but with specific improvements for extension fields. The second method is based on Newton polyhedra, and can be considered to be a generalization of parity sets.

5.1 Dominant trails

Let $F = F_r \circ \dots \circ F_2 \circ F_1$ be a composition of functions. It was shown in Section 3.5 that the correlation A_{χ_{r+1}, χ_1}^F is the sum of the correlations of all ultrametric integral trails of the form $(\chi_1, \chi_2, \dots, \chi_{r+1})$, see (3). The principle of dominant trails is that one can sometimes find a small set of trails Λ so that $A_{\chi_{r+1}, \chi_1}^F \approx \sum_{\chi \in \Lambda} \prod_{i=1}^r A_{\chi_{i+1}, \chi_i}^{F_i}$. By (3), the error is equal to $\sum_{\chi \notin \Lambda} \prod_{i=1}^r A_{\chi_{i+1}, \chi_i}^{F_i}$. If the p -adic absolute value of the error term is less than, say, $1/p^{m+1}$, then the approximation is an equality modulo p^m . In the applications in Sections 6 and 7, the goal is always to show that $A_{\chi_{r+1}, \chi_1}^F \approx 0$. In this case, $\Lambda = \emptyset$. The discussion below focuses on this case, although the methods generalize.

Using the ultrametric triangle inequality $|x + y|_p \leq \max\{|x|_p, |y|_p\}$, the equality (3) in Section 3.5 implies $|A_{\chi_{r+1}, \chi_1}^F|_p \leq \max_{\chi_2, \dots, \chi_r} \prod_{i=1}^r |A_{\chi_{i+1}, \chi_i}^{F_i}|_p$. In particular, to show that A_{χ_{r+1}, χ_1}^F is divisible by p^l , it is sufficient to show that all trails have absolute correlation less than $1/p^l$. In terms of p -adic valuations,

$$\text{ord}_p A_{\chi_{r+1}, \chi_1}^F \geq \min_{\chi_2, \dots, \chi_r} \sum_{i=1}^r \text{ord}_p A_{\chi_{i+1}, \chi_i}^{F_i}.$$

This approach is used for the applications in Section 7.1. However, for extension fields, the models are more difficult to solve. This is due to the greater complexity of the expressions describing addition (Theorems 8 and 9). If the functions F_i are defined over $\mathbf{F}_q$ with $q = p^e$, then the p -adic valuation of $A_{\chi_{i+1}, \chi_i}^{F_i}$ can often be decomposed as follows:

$$\text{ord}_p A_{\chi_{i+1}, \chi_i}^{F_i} = \sum_{j=0}^{e-1} \nu_i(\chi_{i+1}^{p^j}, \chi_i^{p^j}). \quad (6)$$

For example, by Theorem 6, this holds for all monoid homomorphisms (such as monomial functions) and for multiplication by a constant. The equivalent formulation of Theorem 9 in (5) shows that this also applies to addition. Hence, for constructions defined entirely over $\mathbf{F}_q$,

$$\text{ord}_p A_{\chi_{r+1}, \chi_1}^F \geq \min_{\chi_2, \dots, \chi_r} \sum_{j=0}^{e-1} \sum_{i=1}^r \nu_i(\chi_{i+1}^{p^j}, \chi_i^{p^j}) \geq \sum_{j=0}^{e-1} \min_{\chi_2, \dots, \chi_r} \sum_{i=1}^r \nu_i(\chi_{i+1}^{p^j}, \chi_i^{p^j}).$$

This method of bounding trail correlations has two advantages. First, the values ν_i are easier to model using constraint programming or mixed-integer linear programming because they depend linearly on the exponents of the characters $\chi_i^{p^j}$ and $\chi_{i+1}^{p^j}$. Second, as shown in Section 5.2, it makes it easier to represent the set of trails in a compact way.

5.2 Newton polyhedra

If the prime number p is large, then the methods from Section 5.1 can become difficult to apply. For example, most MILP solvers do not support integers greater

than 2^{64} . Similarly, the performance of SMT solvers tends to degrade as p increases. Hence, for some of the applications in Sections 6 and 7, we introduce a coarser but more compact description for the propagation of trails.

In integral cryptanalysis over $\mathbf{F}_2^n$, parity sets [16] are sometimes used to describe properties. The parity set of a set $S \subseteq \mathbf{F}_2^n$ is the support of the modulo-two reduction of $\mathcal{W}(\delta_S)$. To generalize this approach, let w be a vector in $k[X]$ and $\hat{w} = \mathcal{W}(w) = \sum_{\chi} \hat{w}_{\chi} \delta_{\chi}$. If X is a product of n subfields of $\mathbf{F}_q$ with $q = p^e$, then a decomposition similar to (6) is often possible for the p -adic valuations of the coordinates of $\hat{w}$: $\text{ord}_p \hat{w}_{\chi} = \sum_{i=0}^{e-1} \nu(\chi^{p^i})$. The *Newton polyhedron* of w is the convex hull $\overline{\Delta}_w$ of the following set:

$$\Delta_w = \left\{ (u, \nu(\psi)) \mid \psi \in \hat{X} \text{ s.t. } \psi(x) = \tau(x^u) \right\} \subseteq \mathbf{R}^n \times (\mathbf{R} \cup \{\infty\}).$$

Recall that the convex hull of Δ_w is the smallest convex set $\overline{\Delta}_w$ containing Δ_w . The importance of convexity is that it leads to a compact representation of the set as an intersection of half-spaces. The terminology ‘Newton polyhedron’ refers to the *Newton polygon* of polynomials and power series, see for example [28, Section I.4]. The polyhedron $\overline{\Delta}_w$ is infinite, but in practice all points with $\nu(\chi) > m$ may be omitted when correlations are reduced modulo p^m . This results in a finite polygon. Parity sets are an example, since sets of points with $\{0, 1\}$ -coordinates are always convex.

Example 8. Let $w = \delta_{\mathbf{F}_q} \in k[\mathbf{F}_q]$. As shown in Section 3.4, $\hat{w} = q \delta_1 + (q-1) \delta_{\chi}$ with $\chi(x) = \tau(x^{q-1})$. Hence, $\text{ord}_p \hat{w}_{\psi} = \sum_{i=0}^{e-1} \nu(\psi^{p^i})$ with $\nu(\chi) = 0$, $\nu(1) = 1$ and $\nu(\psi) = \infty$ for all $\psi \neq 1, \chi$. The Newton polyhedron $\overline{\Delta}_w$ of w is illustrated in Figure 2. The corresponding polygon for reduction modulo p^2 is shown in blue.

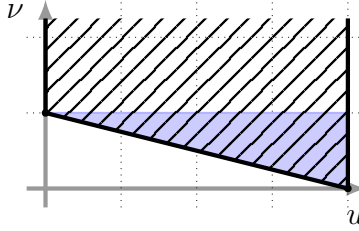


Fig. 2: The polyhedron $\overline{\Delta}_w$ of w (hatched) and modulo- p^2 polygon (blue).

Let $\nu_i(\chi, \psi)$ be defined like in (6). To compute $A^F u$ for a function $F = F_r \circ \dots \circ F_1$, define the polyhedra $\overline{\Delta}_i$ as the convex hulls of the sets

$$\Delta_i = \left\{ (u, v, \nu_i(\chi, \psi)) \mid \psi \in \hat{\mathbf{F}}_q^n \text{ s.t. } \psi(x) = \tau(x^u), \chi \in \hat{\mathbf{F}}_q^m \text{ s.t. } \chi(x) = \tau(x^v) \right\}.$$

The set Δ_w can be combined with Δ_1 as follows:

$$\Delta_w \bowtie \Delta_1 = \left\{ (v, x + y) \mid (u, x) \in \Delta_w \text{ and } (u, v, y) \in \Delta_i \right\}.$$

If two convex sets are combined in this way, then the result is always convex. Furthermore, $\overline{\Delta}_{A^F w} \subseteq \overline{\Delta}_w \bowtie \overline{\Delta}_1$. Similarly, define

$$\Delta_i \bowtie \Delta_{i+1} = \left\{ (u_i, u_{i+2}, x + y) \mid (u_i, u_{i+1}, x) \in \Delta_i \text{ and } (u_{i+1}, u_{i+2}, y) \in \Delta_{i+1} \right\}.$$

By the discussion at the end of Section 5.1, the set $\Delta_i \bowtie \Delta_{i+1}$ corresponds to an upper bound on the p -adic valuations of the composition $F_{i+1} \circ F_i$. Furthermore, this operation is associative. By combining the observations above, one obtains $\overline{\Delta}_{A^F w} \subseteq \overline{\Delta}_w \bowtie \overline{\Delta}_1 \bowtie \overline{\Delta}_2 \bowtie \cdots \bowtie \overline{\Delta}_{r+1}$. As long as the polyhedra $\overline{\Delta}_w, \overline{\Delta}_1, \dots, \overline{\Delta}_{r+1}$ are low-dimensional, the right-hand side above can be computed efficiently using a sequence of affine transformations of polyhedra. Our implementation of this method is based on SAGE support for convex polyhedra [36].

6 Analysis of generic constructions

This section analyzes a number of generic constructions. These results provide designers with a lower bound on the number of rounds required to achieve a certain degree, polynomial density, or to avoid low-data integral properties. They can also serve as a reference point, for instance to compare a specific instance of a construction to the generic case.

6.1 Bounded-degree functions

The constructions analyzed in Sections 6.2 and 6.3 are based on generic functions or permutations of degree d . This section analyzes the properties of such functions. This also serves as a pen-and-paper demonstration of the methods introduced in Section 5.

Theorem 10. *Let $q = p^e$ be a prime power and $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_q^m$ a function of degree d . For all characters ψ of $\mathbf{F}_q^n$ and χ of $\mathbf{F}_q^m$,*

$$\text{ord}_p A_{\chi, \psi}^F \geq \sum_{i=0}^{e-1} \left\lceil \frac{\text{wt}_q \psi^{p^i} / d - \text{wt}_q \chi^{p^i}}{q-1} \right\rceil.$$

Proof. Every coordinate function of F is a linear combination of monomials of degree at most d . A circuit-representation is shown in Figure 3 — note that the constant term can be realized by adding a degree-zero monomial. Since the bound is linear in $\text{wt}_q \chi^{p^i}$ up to rounding, it turns out to be sufficient to prove the result for $m = 1$.

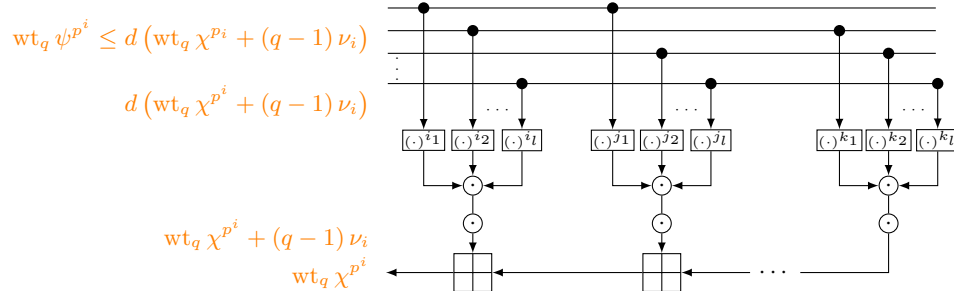


Fig. 3: Ultrametric integral trails for a function with degree d .

The proof strategy is to upper bound the q -weight of the input character of every ultrametric integral trail with output character χ and absolute correlation at most $p^{-\nu}$. To bound trail correlations, the method from Section 5.1 is used. In fact, keeping track of an upper bound on the q -weight of characters can be interpreted as a coarse version of the Newton polyhedra method from Section 5.2.

Let ν be the p -adic valuation of the correlation over the first addition. By Theorem 9 and (5) in particular, the input character ρ of the first addition satisfies $\text{wt}_q \rho^{p^i} \leq \text{wt}_q \chi^{p^i} + (q-1)\nu_i$ for all i in $\{0, 1, \dots, e-1\}$ and with $\nu = \nu_0 + \dots + \nu_{e-1}$. Repeated application shows that this result extends to arbitrary compositions of additions. The absolute correlation for constant multiplications is one, provided that the input and output character are the same. Monomials of degree d are monoid homomorphisms, so the correlation over this step is either zero or one. As discussed in Section 3.5, the q -weight of the input character of the layer of degree- d monomials is at most $d(\text{wt}_q \chi^{p^i} + (q-1)\nu_i)$. The layer of copy operations is again a multiplicative homomorphism and, as discussed in Section 3.5, the q -weight of the input character is always greater than that of the output character — and similarly for all p^{th} powers. Hence,

$$\text{wt}_q \psi^{p^i} \leq d(\text{wt}_q \chi^{p^i} + (q-1)\nu_i). \quad (7)$$

Taking into account that the values ν_i are integers,

$$\nu = \nu_0 + \dots + \nu_{e-1} \geq \sum_{i=0}^{e-1} \left\lceil \frac{\text{wt}_q \psi^{p^i} / d - \text{wt}_q \chi^{p^i}}{q-1} \right\rceil.$$

The result for $m > 1$ follows from the observation that (7) is linear in $\text{wt}_q \psi^{p^i}$, $\text{wt}_q \chi^{p^i}$ and $\nu_0, \dots, \nu_{e-1}$. To complete the argument, introduce additional copy operations at the input and consider trails through each of the m coordinates functions. $\square$

Theorem 10 is used in Sections 6.2 and 6.3 to model generic functions of degree d inside a larger construction. Theorem 10 is also of theoretical interest. For example, it implies the general form of the Ax-Katz theorem [27] using the same argument as in [15, Corollary 6.1].

Remark 2. If F is a permutation, then Theorem 10 can be sharpened. For example, for $n = 1$, this leads to the following additional conditions. A permutation maps $\mathbf{F}_q$ to $\mathbf{F}_q$, so $\delta_{\mathbf{F}_q}$ is an eigenvector of T^F . Since $\delta_{\mathbf{F}_q} = q b_1 + (q-1) b_\psi$ with ψ the character $\psi(x) = \tau(x^{q-1})$, this condition implies $q A_{\chi,1}^F + (q-1) A_{\chi,\psi}^F = 0$ for all characters $\chi \notin \{1, \psi\}$. It follows that $\text{ord}_p A_{\chi,\psi}^F \geq e$. The case $n > 1$ is discussed in Section B.1, but not used in the applications.

Bounding the correlations of individual trails as in Theorem 10 leads to similar results for several other functions. For example, the following result is used in this section and Section 7 to convert between the rings $\mathbf{F}_q^n$ and $\mathbf{F}_{q^n}$. This is useful to reduce the data-complexity of properties.

Theorem 11. Let $\mathbf{F}_{q^n}$ be an extension of a finite field $\mathbf{F}_q$ and $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_{q^n}$ an isomorphism of $\mathbf{F}_q$ -vector spaces. For all characters $\psi = \psi_1 \oplus \dots \oplus \psi_n$ of $\mathbf{F}_q^n$ and χ of $\mathbf{F}_{q^n}$, $A_{\chi, \psi}^F = 0$ unless $\psi_1 \dots \psi_n = \chi|_{\mathbf{F}_q}$ and, if $\chi(x) = \tau(x^{q^n-1})$, ψ is idempotent. Otherwise, for $q = p^e$,

$$\text{ord}_p A_{\chi, \psi}^F \geq \sum_{i=0}^{e-1} \max \left\{ 0, \frac{\text{wt}_q \psi^{p^i} - \text{wt}_q \chi^{p^i}}{q-1} \right\}.$$

Furthermore, $A_{\psi, \chi}^{F^{-1}} = 0$ unless $\psi_1 \dots \psi_n = \chi|_{\mathbf{F}_q}$ and, if $\chi(x) = \tau(x^{q^n-1})$, ψ is idempotent. Otherwise,

$$\text{ord}_p A_{\psi, \chi}^{F^{-1}} \geq \sum_{i=0}^{e-1} \max \left\{ 0, \frac{\text{wt}_q \chi^{p^i} - \text{wt}_q \psi^{p^i}}{q-1} \right\}.$$

Proof. The proof can be found in Section B.2. $\square$

Theorem 11 implies the following variant of Theorem 10 for functions $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_q^m$ with degree d as a function $\mathbf{F}_r^{gn} \rightarrow \mathbf{F}_r^{gm}$ with $\mathbf{F}_r$ a subfield of $\mathbf{F}_q$ for $r = p^f$:

$$\text{ord}_p A_{\chi, \psi}^F \geq \sum_{i=0}^{f-1} \left\lceil \frac{\text{wt}_r \psi^{p^i} / d - \text{wt}_r \chi^{p^i}}{r-1} \right\rceil.$$

The derivation, which amounts to composing F with the isomorphisms from Theorem 11, can be found in Section B.3.

6.2 SHARK- and AES-like constructions

This section analyzes generic SHARK and AES-like construction over a finite field $\mathbf{F}_q$. The round function of SHARK consists of a parallel layer of S-boxes and multiplication with a dense matrix $n \times n$ matrix over $\mathbf{F}_q$. The S-boxes are assumed to be generic permutations of degree d .

Using the method from Section 5.2, we built an automated tool to analyze the ultrametric integral properties of SHARK-like constructions. With minor changes, the same tool can be used for AES-like constructions. The tool computes a compact representation of the Newton polyhedron corresponding to the transition matrix of the function. This representation can be used to extract information such as the degree or minimum-data properties by solving a mixed-integer linear programming problem.⁴

To demonstrate the tool, we choose $n = 8$, $d = 7$ and q as one of $(2^{17} - 1)^4$, $(2^{31} - 2^{24} + 1)^2$ and $2^{64} - 2^{32} + 1$. The degree grows as d^r except that $\lceil \log_d D \rceil + 1$ rather than $\lceil \log_d D \rceil$ rounds are required to reach the maximum degree $D = n(q-1) - 1$. The degree over subfields is more interesting. It can be extracted from the model using Theorem 11, see Figure 4. The degree grows step-wise, a phenomenon that was also observed over $\mathbf{F}_2^n$.

Minimum-data properties, taking into account conversion to a vector space over a subfield and the use of subgroups, are listed in Table 3 in Section B.4.

⁴ The code can be found in `SPN.ipynb` in the git repository.

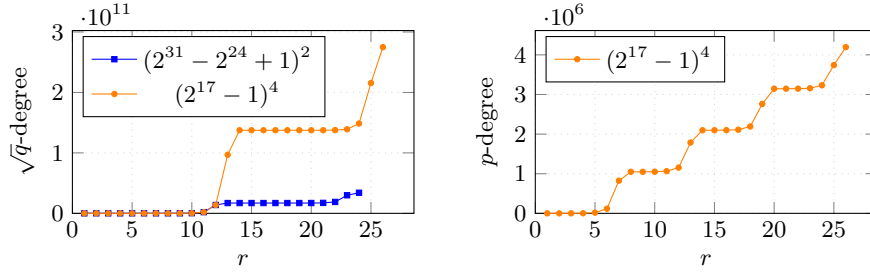


Fig. 4: Degree growth of the SHARK construction with $n = 8$ and $d = 7$.

6.3 Feistel ciphers

For generic Feistel ciphers, the Newton polyhedra method can also be used. This is provided as a part of our tool.⁵ For illustration purposes, we consider a Feistel cipher with a generic degree $d = 3$ round function over $\mathbf{F}_q$ with q equal to $(2^{17} - 1)^4$, $(2^{31} - 2^{24} + 1)^2$ or $2^{64} - 2^{32} + 1$.

For $q = 2^{64} - 2^{32} + 1$, the degree is shown in Figure 5. As the number of r rounds increases, it first grows exponentially as d^r until $r = s$ with $d^s \approx q$. For the remaining rounds, the degree increases slowly as $2(q - 1) - d^{s-r}$. Similar graphs for the other choices of q can be found in Section B.5. By counting lattice points in a polygon using SAGE [36], the Newton polyhedron method can be used to compute the fraction of monomials that occur in the algebraic normal form of the coordinate functions. An example is shown in Figure 5; the density is expressed as a fraction of the total number of monomials of degree up to the Feistel cipher degree.

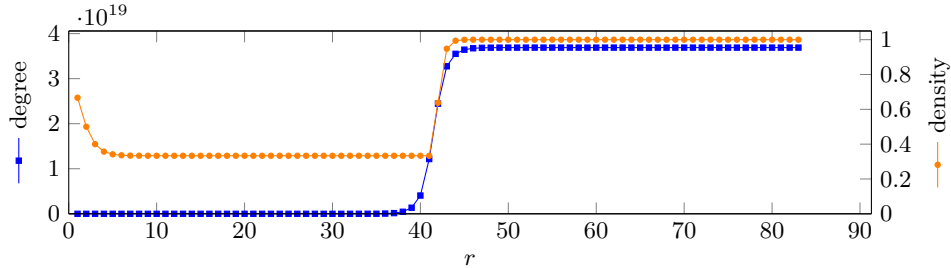


Fig. 5: Degree and density for Feistel ciphers with $d = 3$ and $q = 2^{64} - 2^{32} + 1$.

Minimal-data properties for the three examples mentioned above are listed in Section B.5. Like for the SHARK construction, data-complexity can sometimes be reduced by exploiting submonoids.

7 Concrete applications

This section applies the methods from Section 5 to concrete primitives. The main difference with Section 6 is that the S-boxes used in these primitives are

⁵ See `Feistel.ipynb` in the git repository.

power functions $x \mapsto x^d$. As discussed at the end of Section 3.5, these are monoid homomorphisms.

7.1 Primitives defined over small fields

For primitives defined over a product of small fields, exact modelling of the p -adic valuations of trail correlations is feasible. The method from Section 5.2 based on Newton polyhedra is not necessary to analyze these primitives. This section illustrates this with results on AES-prime, small-pSquare, and mid-pSquare [6, 23, 33]. These ciphers are defined over $\mathbf{F}_p^n$, with $p \leq 2^{32}$. The tool we have developed can also be applied to ciphers over similarly small fields.⁶

AES-prime. The complete specification of AES-prime can be found in the Eurocrypt paper by Masure et al. [33]. It is an AES-like cipher, with a 4×4 state of $\mathbf{F}_p$ elements. The S-boxes are functions of the form $x \mapsto x^d + c$, with c a constant. Concretely, for $p = 2^7 - 1$, the authors propose $d = 5$ and $c = 2$. The MDS-matrix behaves like a generic linear permutation.

Degree growth. The designers argue that the degree of AES-prime grows as d^r with r the number of rounds, until the maximum degree of $16 \cdot (p-1) - 1 = 2015$ is reached. Our analysis reveals that this is not true. The degree growth is shown in Figure 6. Here, the concrete model agrees with the generic analysis from Section 6.2. However, AES-prime does *not* behave like a generic AES-like cipher. For example, Figure 6 also shows the ‘modulo- p^2 degree’, which is lower than for the generic case and takes twice as many rounds to reach the maximum of 2016. This is the maximum value of $\text{wt}_p \psi$ so that $A_{\chi, \psi}^F \not\equiv 0 \pmod{p^2}$ for $\text{wt}_p \chi = 1$.

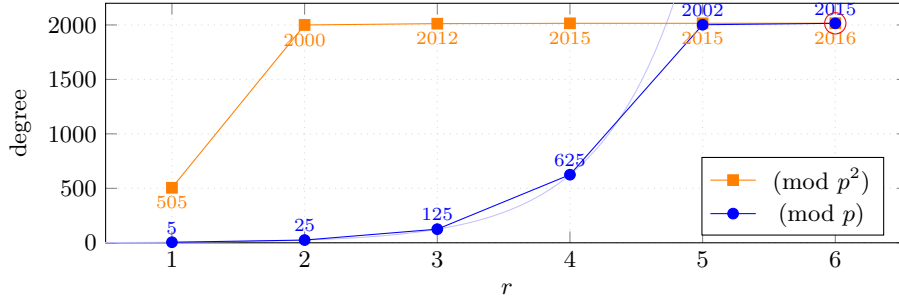


Fig. 6: Degree growth of AES-prime in terms of the number of rounds r .

Density of the algebraic normal form. The number of rounds of AES-prime is determined by the security argument against interpolation attacks. It is argued that the algebraic normal form of all coordinate functions should be completely dense to avoid this attack. This requires 10 rounds according to the authors [33, page 19]. Our analysis suggests that seven rounds are sufficient. Hence, the conclusion that interpolation attacks are dominant is invalid. The next paragraph shows that there are better attack strategies.

⁶ See `AESprime.ipynb`, `small-psquare.ipynb` and `mid-psquare.ipynb` of the git repository.

Integral properties. As mentioned above, there is a four round zero-sum property with p data. By saturating one diagonal, a five round property with p^4 data is obtained. Saturating two diagonals yields a zero-sum property on six rounds with p^8 data. Higher divisibility properties on **AES-prime** require significantly more data. For 3.5 rounds, there are zero-sums modulo p^2 with p^8 data, modulo p^3 with p^{12} data and modulo p^4 with p^{15} data.

All of these properties can be extended to key-recovery attacks on two more rounds by guessing five words of the key using the FFT method [39], for a time-complexity of around $28p^5$ arithmetic operations in $\mathbf{F}_p$ per data set. Six data sets are sufficient. This results in an eight-round attack with $6p^8 \leq 2^{59}$ data and $6 \times 28p^5 \leq 2^{43}$ arithmetic operations to recover five key words. This is better than for the AES, and improves over the attacks discussed by the designers.

small-pSquare. The tweakable block cipher **small-pSquare** is a generalized Feistel cipher of type II, with b branches in $\mathbf{F}_p^c$. The Feistel round function is shown in Figure 7. Concretely, the authors propose $p = 2^7 - 1$ and $b = c = 4$. The analysis below focuses on this instance. For the complete **small-pSquare** specification, we refer to the Eurocrypt paper by Grassi et al. [23].

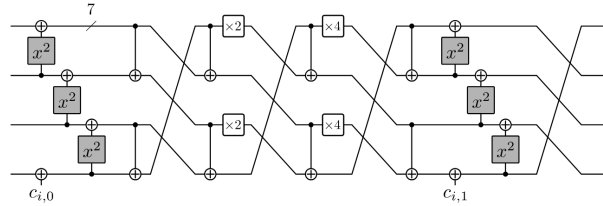


Fig. 7: The Feistel round function of **small-pSquare** for $c = 4$. Figure from [23].

Degree growth and density of the algebraic normal form. The designers argue that the degree of **small-pSquare** grows as 4^r until reaching the maximum of $16 \cdot (p - 1) - 1$ after six rounds. The minimum degree of all coordinate functions is claimed to be $4^r/8$, reaching the maximum degree after seven rounds. Our analysis shows that 13 rounds are necessary to reach the maximum degree. Moreover, 15 rounds are necessary for all coordinate functions to reach the maximum degree. Detailed results are shown in Figure 8.

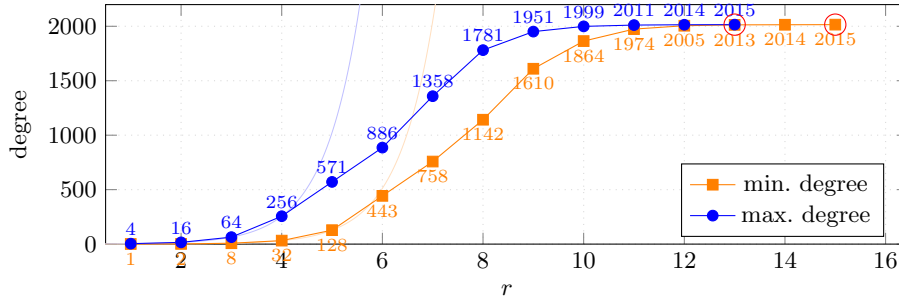


Fig. 8: Degree growth of **small-pSquare** in terms of the number of rounds r .

To determine the impact of interpolation attacks, the designers of **small-pSquare** estimate that 12 rounds are required for the algebraic normal form of every coordinate function to be dense. We show that 17 rounds are required instead. They argue that $2r + 4$ rounds are sufficient if full density is reached after r rounds [23, §D.3]. Since $2r + 4 = 38$, this is more than the proposed number of rounds (36) for instances without tweak. Hence, **small-pSquare** does not meet its design criteria. It is unlikely that an interpolation attack against the full-round instance can actually be mounted, because the analysis in [23, §D.3] is too optimistic. These observations question the claim that “the main attack vector against **small-pSquare** is differential cryptanalysis” [23, §5].

Integral properties. Integral properties of **small-pSquare** that contradict the analysis of higher-order differentials in [23, §D.3] can be found in Section C.1.

mid-pSquare. The tweakable block cipher family **mid-pSquare** is based on the same design strategy as **small-pSquare**, but using a larger prime p . Concretely, for the instance proposed by the designers, $p = 2^{31} - 1$ and $b = c = 2$. Additional details about **mid-pSquare** can be found in [6].

Degree growth and density of the algebraic normal form. The security analysis of **mid-pSquare** is similar to that of **small-pSquare**. The designers argue that 20 rounds are sufficient to attain the maximum degree in each coordinate function. We show 35 rounds are necessary. The designers estimate that after 20 rounds one coordinate function is dense, and after 24 rounds every coordinate function is dense. We show this should be modified to 35 and 37 rounds respectively.

For interpolation attacks, the designers argue that $2r + 2 + 8\tau$ rounds are necessary [6, §3.2.3]. Here, r is the number of rounds required to achieve full degree and τ is the tweak length. Since $r = 37$, this means $76 + 8\tau$ rounds are necessary according to the criteria proposed by the designers. This exceeds the proposed number of rounds, i.e. $56 + 8\tau$.

Integral properties. For **mid-pSquare**, nontrivial zero-sum properties exist for up to 36 rounds when using the full codebook. For 21 rounds, there is an integral property that can be tested using $3p^3 \leq 2^{95}$ data. This property is based on the submonoid of inputs $\{0, 1, p - 1\} \times \mathbf{F}_p \times \mathbf{F}_p \times \mathbf{F}_p$, but with unequal weights. As explained in Section 3.4, the input state is chosen as $(\delta_{\{1, p-1\}} - 2\delta_0) \otimes \delta_{\mathbf{F}_p} \otimes \delta_{\mathbf{F}_p} \otimes \delta_{\mathbf{F}_p}$. For higher-divisibility properties, zero-sums modulo p^2 exist for up to 19 rounds. Divisibility by p^3 is possible for at most two rounds.

7.2 HadesMiMC and GMiMC

For primitives defined over fields with a large characteristic, the Newton polyhedra method from Section 5.2 can be used.⁷

⁷ These cases are included in the tool, see `HadesMiMC-Rf.ipynb` and `HadesMiMC-Rp.ipynb` of the git repository.

HadesMiMC. HadesMiMC [22] is a keyed permutation that combines full SHARK-like rounds with partial rounds with one S-box. The purpose of the partial rounds is to increase the degree of the cipher while minimizing the number of required multiplications. The S-boxes are instantiated with the power map $x \mapsto x^d$.

Degree growth. The models from Section 6.2 can be instantiated with the Newton polyhedron for the power map. For the fields and block sizes used in Section 6.2, and for $d = 7$, the degree of HadesMiMC for $\lceil \log_d D \rceil$ rounds, where $D = n(q - 1) - 1$, is lower than in the generic case. Details can be found in Section D.

The designers of HadesMiMC argue security against algebraic attacks purely based on the partial rounds. Interpolation is claimed to be dominant, and the minimum number of rounds required to thwart it is $r_f + r_i + 2$, with r_f and r_i the minimum number of partial and inverse partial rounds with maximal degree and density. The designers estimate $r_f = 1 + \lceil \log_d D \rceil$ and $r_i = 2$.

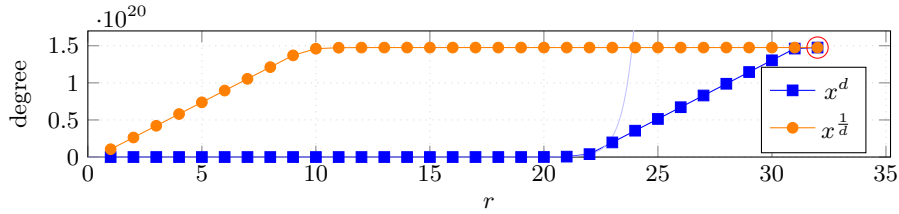


Fig. 9: Degree growth of partial and inverse partial rounds of HadesMiMC with $p = 2^{64} - 2^{32} + 1$ and $d = 7$ in terms of the number of rounds r .

Figure 9 shows the growth of the degree for both partial and inverse partial rounds. It first grows exponentially as d^r for $\lfloor \log_d(q - 1) \rfloor$ rounds, then linearly with slope $(d - 1)\lfloor (q - 1)/d \rfloor$ up to the maximum degree. Figure 9 also shows that partial and inverse partial rounds reach maximal degree after the same number of rounds. Hence, for small d , HadesMiMC requires at least $2(\lfloor \log_d(q - 1) \rfloor + \lceil dt/(d - 1) \rceil + 1)$ rounds to meet its design criteria. This is higher than the bound used by the designers. For example, 88 partial rounds are proposed for a state size and security level of 256 bits over a field of 128 bits, with $d = 3$ [22, Table 1]. The improved degree bounds increase this to 168 rounds.

Feistel-GMiMC. GMiMC [1] is a design strategy for keyed permutations based on generalized Feistel ciphers with round functions of the form $x \mapsto (x + k)^3$. For simplicity, we focus on the two-branch variants. In practice, $x \mapsto (x + k)^3$ behaves like a generic function of degree 3. Therefore, the results from Section 6.3 are directly applicable.

Degree growth. The designers use exponential degree bounds for Feistel-GMiMC. Based on the results in Section 6.3, this underestimates the number of rounds required to reach maximal degree by a factor of two. Therefore, like for HadesMiMC, the minimum number of rounds to protect against interpolation attacks should be approximately doubled. Since the interpolation attack is dominant, this shows

that Feistel-GMiMC does not comply with its own security requirements. For example, 84 rounds are proposed for $p = 2^{64} - 2^{32} + 1$. By plugging the new degree bounds into the arguments of the designers, this should be 169 rounds.

Acknowledgments

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A Supplementary material for Section 3

This appendix contains supplementary material for Section 3.

A.1 Example of Theorem 4

The set $\{1, 4\}$ is a subgroup of $\mathbf{F}_5^\times$. Its indicator $\delta_{\{1,4\}}$ satisfies

$$\mathcal{U}_{\mathbf{F}_5}(\delta_{\{1,4\}}) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & i & -i & -1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 1 & -i & i & -1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 2 \\ 0 \\ 2 \end{bmatrix} = 2\delta_{\{1,\chi,\chi^2\}},$$

where the character $\chi: \mathbf{F}_q \rightarrow k$ is given by $\chi(x) = \tau(x^2)$. In practice, it is better to choose $u = \delta_{\{1,4\}} - 2\delta_0$ since

$$\mathcal{U}_{\mathbf{F}_5}(\delta_{\{1,4\}} - 2\delta_0) = 2\delta_{\{\chi,\chi^2\}}.$$

Importantly, the support of the result does not contain the trivial character.

A.2 Examples of transition matrices

Let $F: \mathbf{F}_5 \rightarrow \mathbf{F}_5$ be the function defined by $F(x) = -x$. The transition matrix A^F is given by

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & i & -i & -1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 1 & -i & i & -1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 & 0 & -4 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & -i & -1 & i & 1 \\ 0 & i & -1 & -i & 1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

The matrix A^F is diagonal with coordinates $A_{\chi,\chi}^F = \chi(-1)$. This is true by construction of the basis, since b_χ is an eigenvector of A^F with eigenvalue $\chi(-1)$. As another example, consider the function $G: \mathbf{F}_5 \rightarrow \mathbf{F}_5$ defined by $x \mapsto x^2$. The matrix A^G is given by

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & i & -i & -1 \\ 0 & 1 & -1 & -1 & 1 \\ 0 & 1 & -i & i & -1 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \frac{1}{4} \begin{bmatrix} 4 & 0 & 0 & 0 & -4 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & -i & -1 & i & 1 \\ 0 & i & -1 & -i & 1 \\ 0 & -1 & 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

B Supplementary material for Section 6

This appendix contains supplementary material for Section 6.

B.1 Generalization of Remark 2

Let $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_q^n$ be a permutation; the case $n = 1$ was discussed in Remark 2. A permutation maps $\mathbf{F}_q^n$ to $\mathbf{F}_q^n$, so $\delta_{\mathbf{F}_q^n}$ is an eigenvector of T^F . Evaluating $b^\chi(T^F \delta_{\mathbf{F}_q^n}) = b^\chi(\delta_{\mathbf{F}_q^n})$ with χ a non-idempotent character yields

$$\sum_{\psi} q^{n-\text{wt } \psi} (q-1)^{\text{wt } \psi} A_{\chi, \psi}^F = 0,$$

where the sum is over all idempotent characters ψ . Let ρ be the maximal degree character $\rho(x) = \tau(\prod_{i=1}^n x_i^{q-1})$. Isolating the term $A_{\chi, \rho}^F$ and taking the p -adic absolute value of both sides yields

$$\text{ord}_p A_{\chi, \rho}^F \geq \min_{\psi} e(n - \text{wt } \psi) + \text{ord}_p A_{\chi, \psi}^F,$$

where the minimum is over all idempotent characters $\psi \neq \rho$. In general, the right-hand side cannot be simplified further without making assumptions about χ . However, if $d \geq n-1$, then $\text{ord}_p A_{\chi, \psi}^F = 0$ and $\text{ord}_p A_{\chi, \rho}^F \geq e$. Hence, permutations of degree $d \geq n-1$ additionally satisfy

$$\text{ord}_p A_{\chi, \rho}^F \geq e,$$

for all non-idempotent characters χ .

B.2 Proof of Theorem 11

Let $\beta_1, \dots, \beta_n$ be a basis for $\mathbf{F}_{q^n}$ as a vector space over $\mathbf{F}_q$. The isomorphism $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_{q^n}$ is of the form

$$F(x) = \sum_{i=1}^n i(x_i) \beta_i.$$

Here, $i: \mathbf{F}_q \rightarrow \mathbf{F}_{q^n}$ is the inclusion homomorphism. By Theorem 6,

$$A_{\chi, \psi}^i = \begin{cases} 1 & \text{if } \chi|_{\mathbf{F}_q} = \psi, \\ 0 & \text{else.} \end{cases}$$

To analyze the function F , we consider ultrametric integral trails through the circuit representation of F in Figure 10. For the addition operation, it must hold that $\chi = \rho_1 \rho_2 \cdots \rho_n$. Hence,

$$\chi|_{\mathbf{F}_q} = \rho_1|_{\mathbf{F}_q} \rho_2|_{\mathbf{F}_q} \cdots \rho_n|_{\mathbf{F}_q} = \psi_1 \psi_2 \cdots \psi_n.$$

If $\chi(x) = \tau(x^{q^n-1})$, then ψ must be idempotent. The reason for this is that, in this case, χ is the indicator function of $\mathbf{F}_{q^n}^\times$. The function F maps $\mathbf{F}_q^n \setminus \{0\}$ to $\mathbf{F}_{q^n}^\times$.

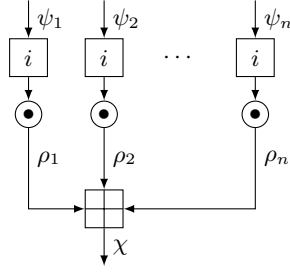


Fig. 10: Circuit representation of F .

The indicator function of the former set is a linear combination of idempotent characters. By Theorem 9, the correlation of the trail in Figure 10 is equal to

$$\sum_{i=0}^{e-1} \frac{\text{wt}_q \rho^{p^i} - \text{wt}_q \chi^{p^i}}{q-1}.$$

Since $\rho_j|_{\mathbf{F}_q} = \psi_j$ for all j in $\{1, \dots, n\}$, we have $\text{wt}_q \rho^{p^i} \geq \text{wt}_q \psi^{p^i}$. The result follows by taking into account that each term in the expression above must be non-negative.

For the inverse of F , a similar argument can be used. Alternatively, it is possible to exploit the relation between A^F and its inverse that follows from the fact that $T^{F^{-1}}$ is the transpose of T^F :

$$A^{F^{-1}} = (\mathcal{U}_{\mathbf{F}_q^n} \mathcal{U}_{\mathbf{F}_q^n}^\top) A^{F^\top} (\mathcal{U}_{\mathbf{F}_{q^n}} \mathcal{U}_{\mathbf{F}_{q^n}}^\top)^{-1}$$

Although it is not true that the inverse of A^F is equal to the transpose of A^F , one can verify that — for valuations — it is close enough to deduce Theorem 11.

B.3 Proof of subfield variant of Theorem 10

Let $F: \mathbf{F}_q^n \rightarrow \mathbf{F}_q^m$ be a function with degree d as a function $\mathbf{F}_r^{gn} \rightarrow \mathbf{F}_r^{gm}$ with $\mathbf{F}_r$ a subfield of $\mathbf{F}_q$, where $r = p^f$:

$$\text{ord}_p A_{\chi, \psi}^F \geq \sum_{i=0}^{f-1} \left\lceil \frac{\text{wt}_r \psi^{p^i}/d - \text{wt}_r \chi^{p^i}}{r-1} \right\rceil.$$

If $G: \mathbf{F}_q \rightarrow \mathbf{F}_r^g$ is the isomorphism from Theorem 11, then

$$\text{ord}_p A_{\chi, \psi}^F \geq \min_{\rho, \lambda} \sum_{i=1}^m \text{ord}_p A_{\chi_i, \lambda_i}^G + \text{ord}_p A_{\lambda, \rho}^F + \sum_{i=1}^n \text{ord}_p A_{\rho_i, \psi_i}^{G^{-1}}.$$

The expression on the right-hand side is bounded from below by

$$\sum_{i=0}^{f-1} \max \left\{ 0, \frac{\text{wt}_r \rho^{p^i} - \text{wt}_r \chi^{p^i}}{r-1} \right\} + \left\lceil \frac{\text{wt}_r \lambda^{p^i}/d - \text{wt}_r \rho^{p^i}}{r-1} \right\rceil \\ + \max \left\{ 0, \frac{\text{wt}_r \psi^{p^i} - \text{wt}_r \lambda^{p^i}}{r-1} \right\}.$$

For $\text{wt}_r \rho^{p^i} \geq \text{wt}_r \chi^{p^i}$ and $\text{wt}_r \lambda^{p^i} \leq \text{wt}_r \psi^{p^i}$, the bound is increasing in $\text{wt}_q \rho^{p^i}$ and decreasing in $\text{wt}_q \lambda^{p^i}$. Hence, the minimum is achieved for $\rho = \chi$ and $\lambda = \psi$.

B.4 SHARK structure

The properties in Table 3 take into account conversion to a vector space over a subfield, and the use of subgroups. This can depend on q , for example, the 24-round property for $q = (2^{31} - 2^{24} + 1)^2$ with $q^7/4$ data is based on the existence of a submonoid of $\mathbf{F}_q$ of order $q/4$ — the union of $\{0\}$ and the cyclic subgroup of order $(q-1)/4$. This property requires weighting the inputs as described in Section 3.4.

The trend in Table 3 is that, for divisibility by p , bigger fields lead to stronger properties. For divisibility by p^2 , the extension degree is more important.

Table 3: Integral properties of the SHARK structure with $n = 8$ and $d = 7$.

q	r (divisibility by p)					
	≤ 21	22	23	24	25	26
$(2^{31} - 2^{24} + 1)^2$	$< q^\dagger$	q	q	$\frac{1}{4}q^7$	—	—
$2^{64} - 2^{32} + 1$	$< q^\dagger$	$< q^\dagger$	q	q^2	—	—
$(2^{17} - 1)^4$	$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	q	q	q^5
	r (divisibility by p^2)					
	1	≤ 12	13	$\dots$	≤ 19	20
$2^{64} - 2^{32} + 1$	q	—	—		—	—
$(2^{31} - 2^{24} + 1)^2$	q	q	q^7		—	—
$(2^{17} - 1)^4$	q	q	q		q	q^7

[†] Data-complexities below q are achieved using submonoids of $\mathbf{F}_q$.

B.5 Feistel ciphers

This appendix contains supplementary material to Section 6.3. Figure 11 depicts the q -degree growth for all three fields. Figure 12 depict the subfield degree growth of the two extension fields. Table 4 contains more properties modulo p and p^2 for all three fields.

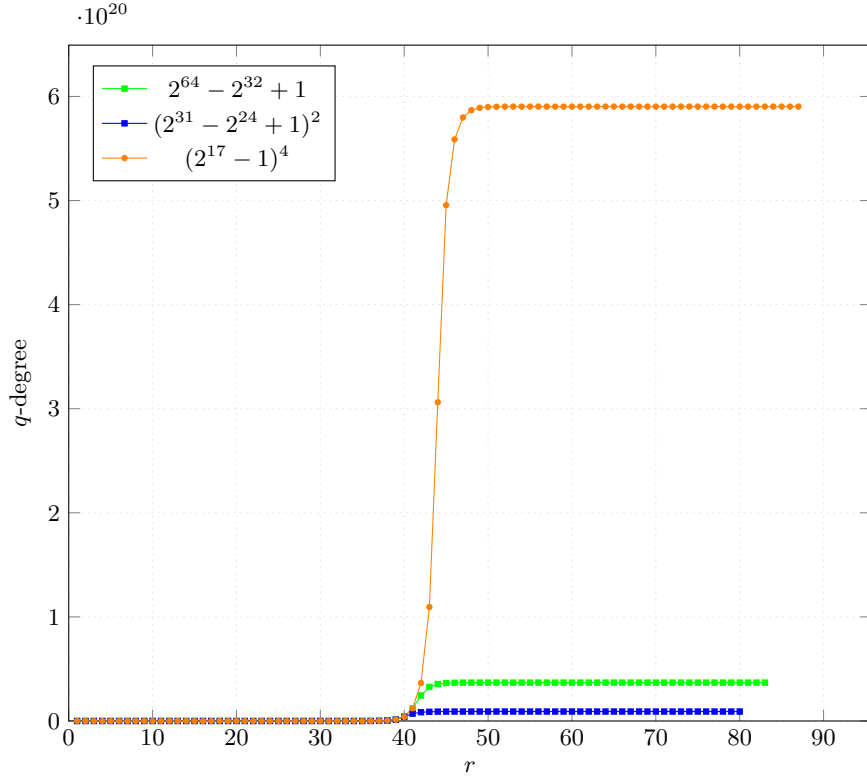


Fig. 11: q -Degree growth of Feistel ciphers with $d = 3$.

C Results for AES-prime, small-pSquare and mid-pSquare

This appendix contains additional results for Section 7.1.

C.1 Integral properties of small-pSquare

Based on the degree bounds in Figure 8, zero-sum properties with p^{16} data can be found for up to 14 rounds. This contradicts the analysis of higher-order differential attacks in [23, §D.3], where it is claimed that 12 rounds are sufficient. For 13 rounds, there exist zero-sums with $p^{15} \leq 2^{105}$ data.

Higher-divisibility properties can be found for up to 11 rounds, for divisibility by p^2 . For six rounds, there exist zero-sums modulo p^3 . All of these properties can be extended to key-recovery attacks on round-reduced instances, but these are unlikely to be dominant for small-pSquare. Hence, the details are not discussed.

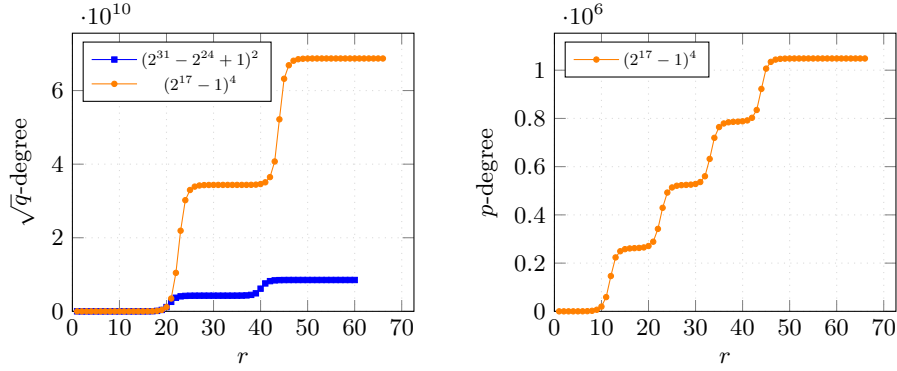


Fig. 12: Degree growth of Feistel ciphers with $d = 3$.

Table 4: Integral properties of Feistel ciphers with $d = 3$.

q	r (divisibility by p)						≤ 79	≤ 82	≤ 83
	≤ 40	41	42	43	44	45			
$(2^{31} - 2^{24} + 1)^2$	$< q^\dagger$	q^2	q^2	q^2	q^2	q^2	q^2	—	—
$2^{64} - 2^{32} + 1$	$< q^\dagger$	$< q^\dagger$	q	$\frac{1}{3}q^2$	q^2	q^2	q^2	q^2	—
$(2^{17} - 1)^4$	$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	$\frac{1}{26}q^2$	q^2	q^2	q^2
r (divisibility by p^2)									
		2	≤ 20	21	≤ 33	34	≤ 40	≤ 65	
$2^{64} - 2^{32} + 1$		$5q$	—	—	—	—	—	—	
$(2^{31} - 2^{24} + 1)^2$		$< q^\dagger$	$< q^\dagger$	q	$< q^2$	$< q^2$	$< q^2$	—	
$(2^{17} - 1)^4$		$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	$< q^\dagger$	q	$< q^2$	$< q^2$	

[†] Data-complexities below q are achieved using submonoids of $\mathbf{F}_q$.

[‡] Data-complexities below q are achieved using submonoids of $\mathbf{F}_p^e$ with $q = p^e$.

C.2 Degree growth in mid-pSquare

The degree of each of the four coordinates is shown in Figure 13. Observe that the degree does not become maximal until round 34 (for two components) or round 35 (for all components).

D Full HadesMiMC versus generic rounds

This appendix is supplementary material to Section 7.2. Table 5 contains the changes in degree between the generic SPN model and full HadesMiMC rounds.

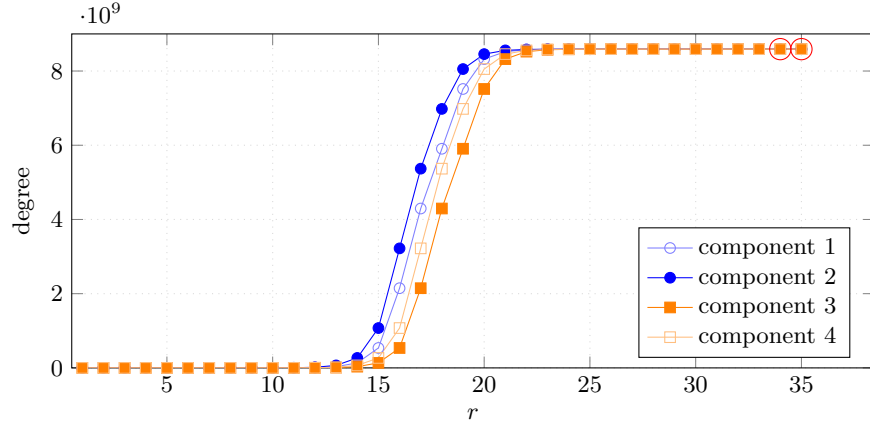


Fig. 13: Degree growth of mid-pSquare in terms of the number of rounds r .

Table 5: Changes in q -degree between r rounds of the examples of Section 6.2 and the same examples instantiated with a power map of the same degree. Δ is the degree of the former minus the degree of the latter.

q	r	Δ
$2^{64} - 2^{32} + 1$	24	28
$(2^{31} - 2^{24} + 1)^2$	24	4
$(2^{17} - 1)^4$	26	8